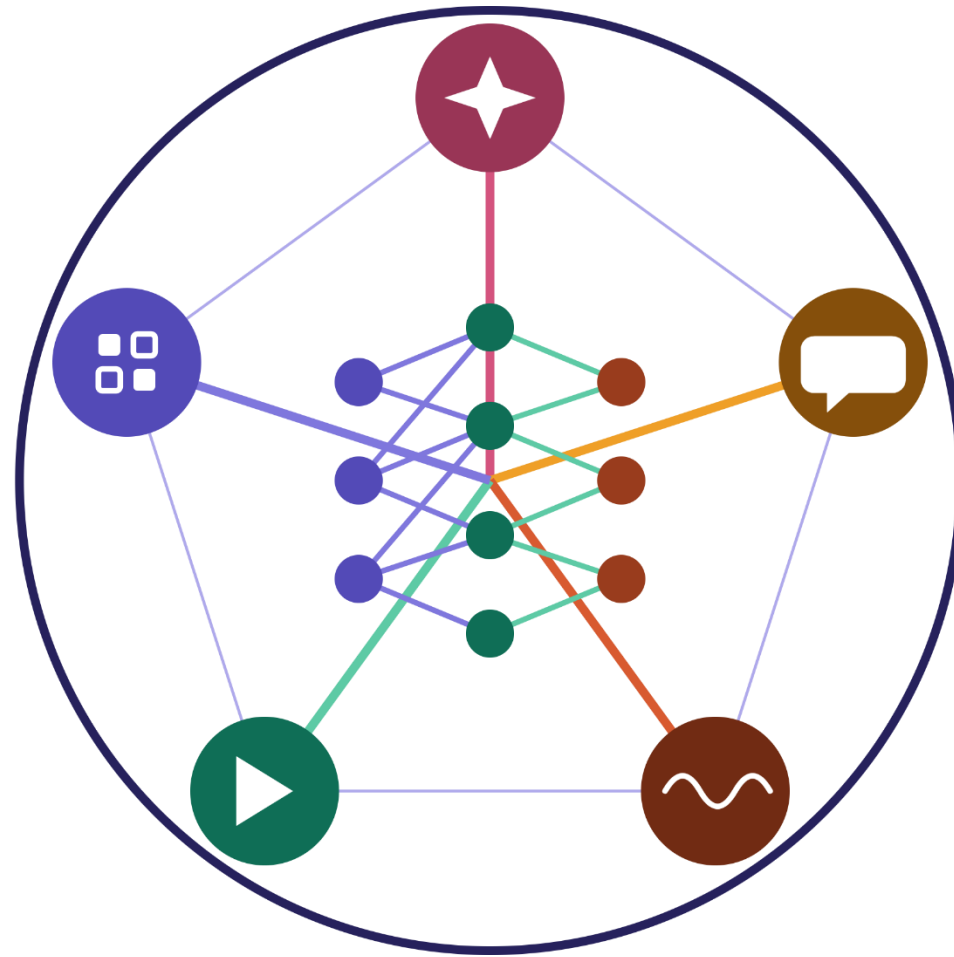




Lecture 01

Machine Learning And Deep Learning An Overview.

Lecture Topics

- 1. Course overview**
2. What is machine learning?
3. The broad categories of ML
4. The supervised learning workflow
5. Necessary ML notation and jargon
6. About the practical aspects and tools

A short teaser: what you
will be able to do after
this course

Audio Classification Using Convolutional Neural Networks

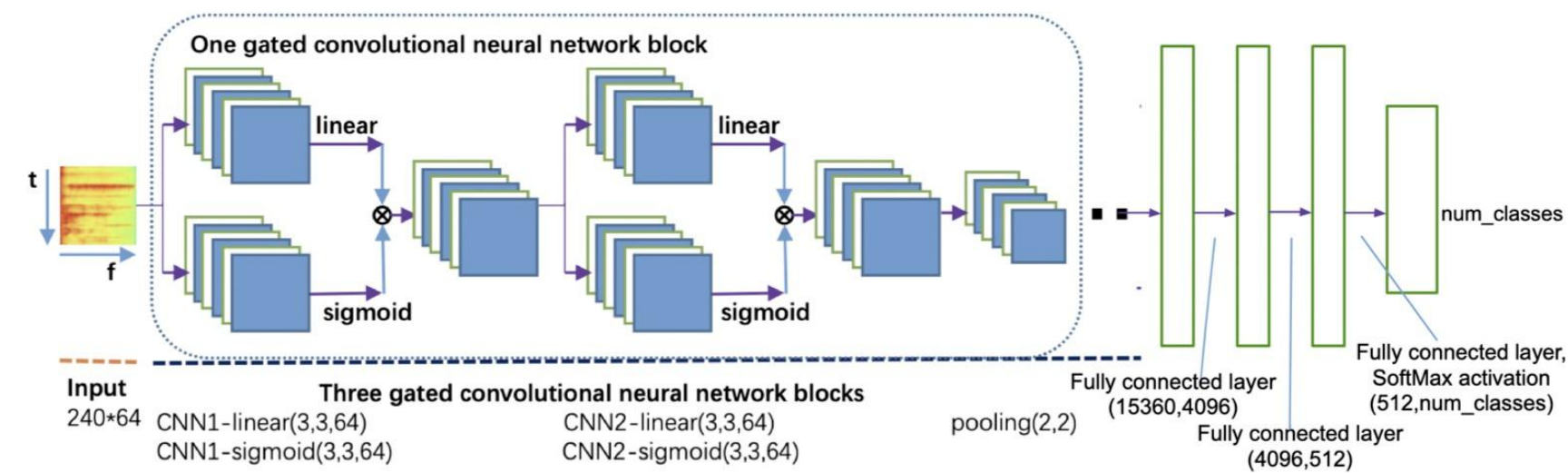
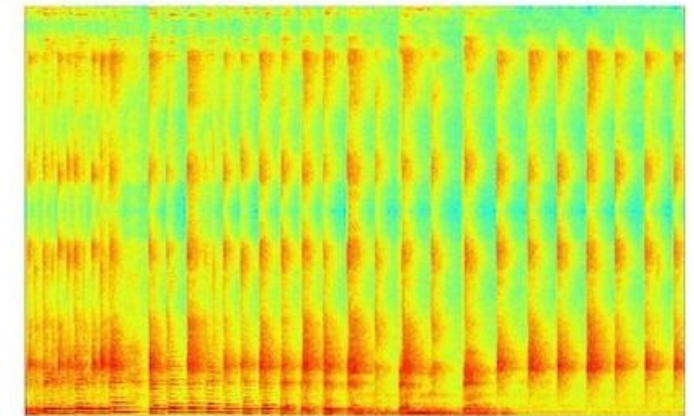
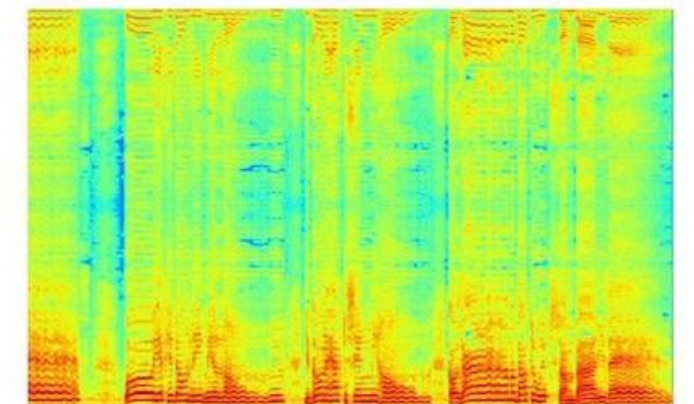


Figure 2. Gated Convolutional Neural Network with Multi-Layered Classification Model

Spectrogram of an audio clip corresponding to "finger snapping:"



Spectrogram of an audio clip corresponding to "synthetic singing:"



3D Convolutional Networks

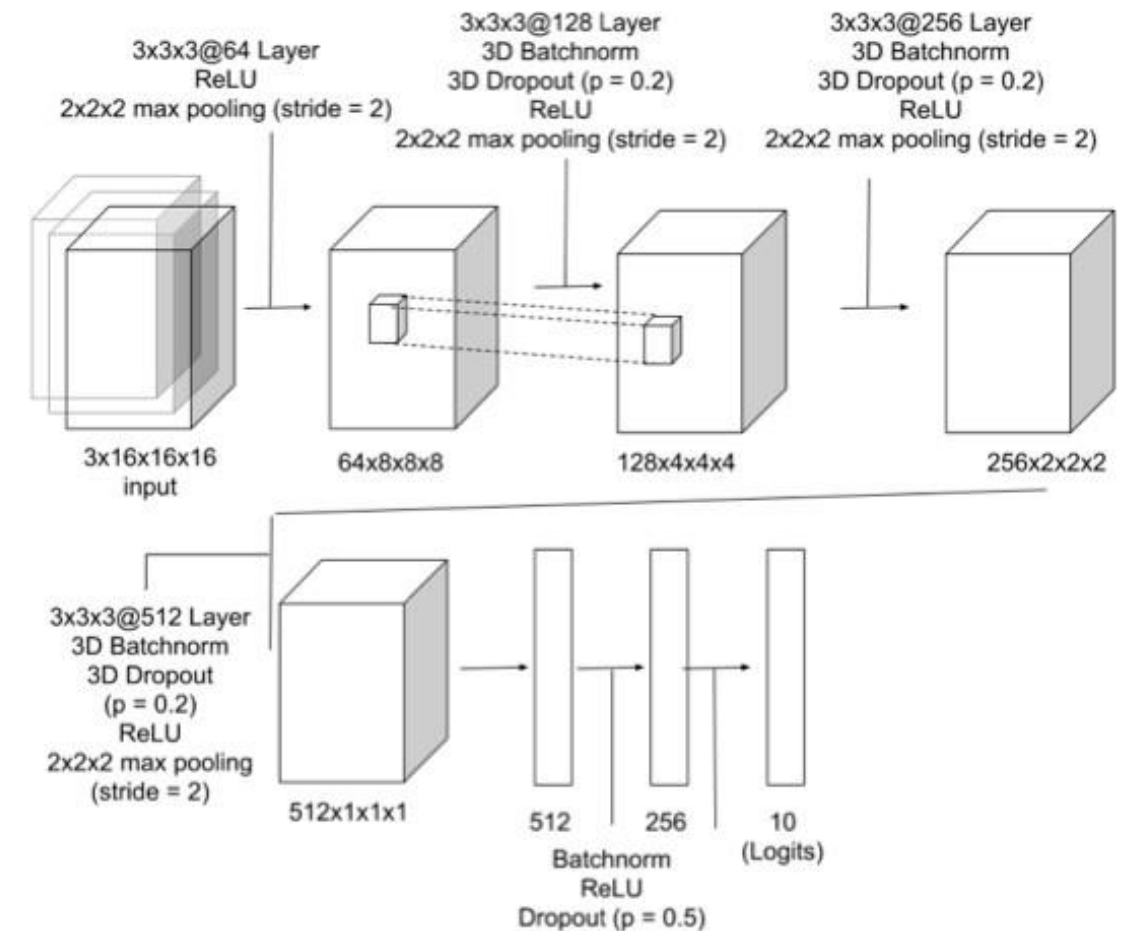
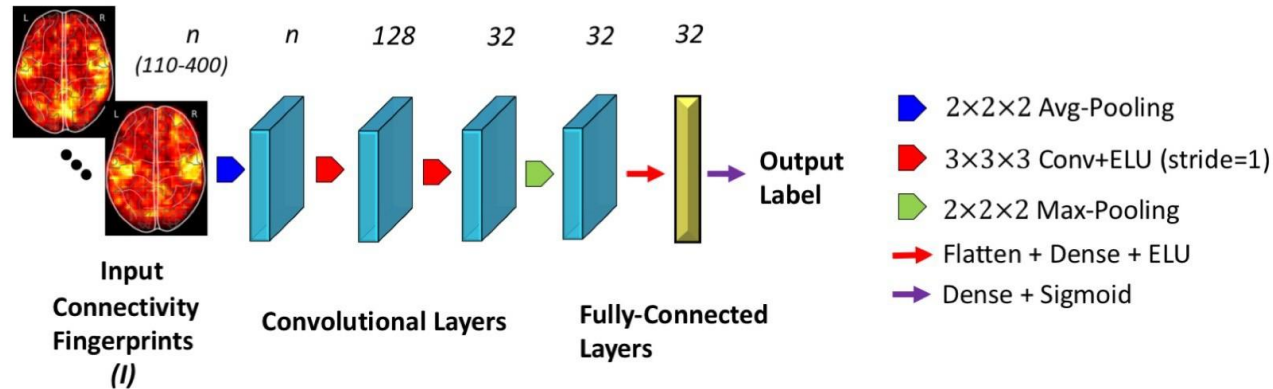
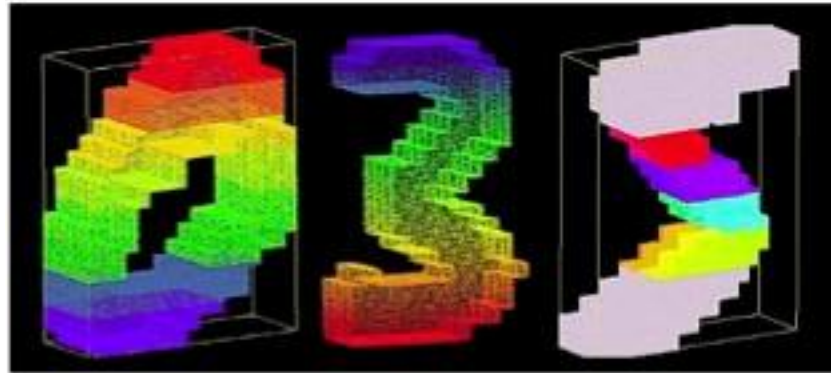


Image Source: 3D Convolutional Neural Networks for Classification of Functional Connectomes. <https://arxiv.org/abs/1806.04209>

Photographic Style Transfer With Deep Learning



Guernica



Model Result



Sandstone



Model Result



Water Drop



Model Result



Skrik



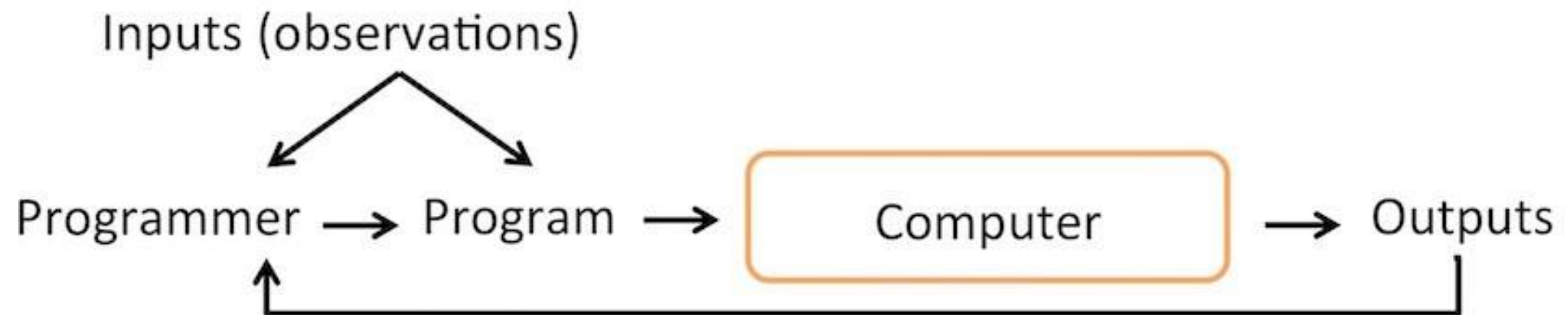
Model Result

What Is Machine Learning?

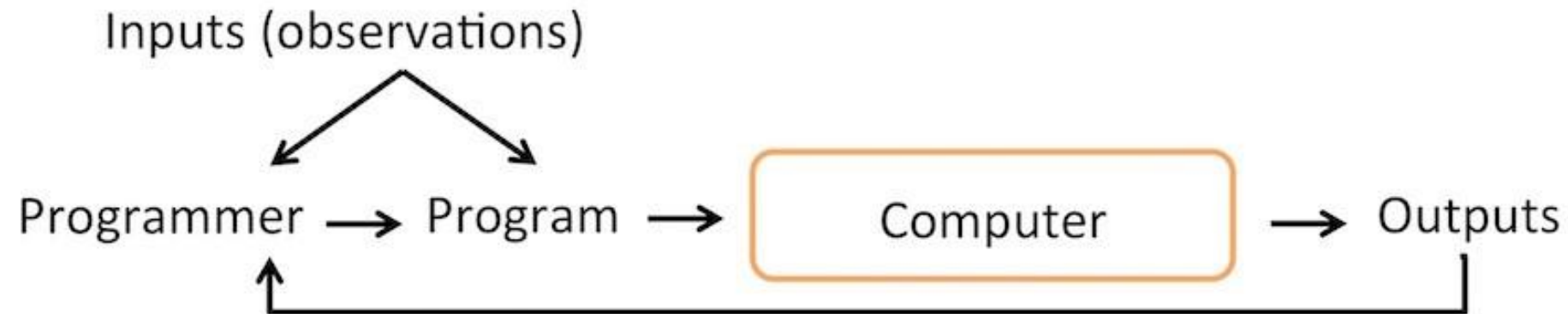
A short overview before we jump into Deep Learning

1. Course overview
- 2. What is machine learning?**
3. The broad categories of ML
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6. About the practical aspects and tools

The Traditional Programming Paradigm



The Traditional Programming Paradigm

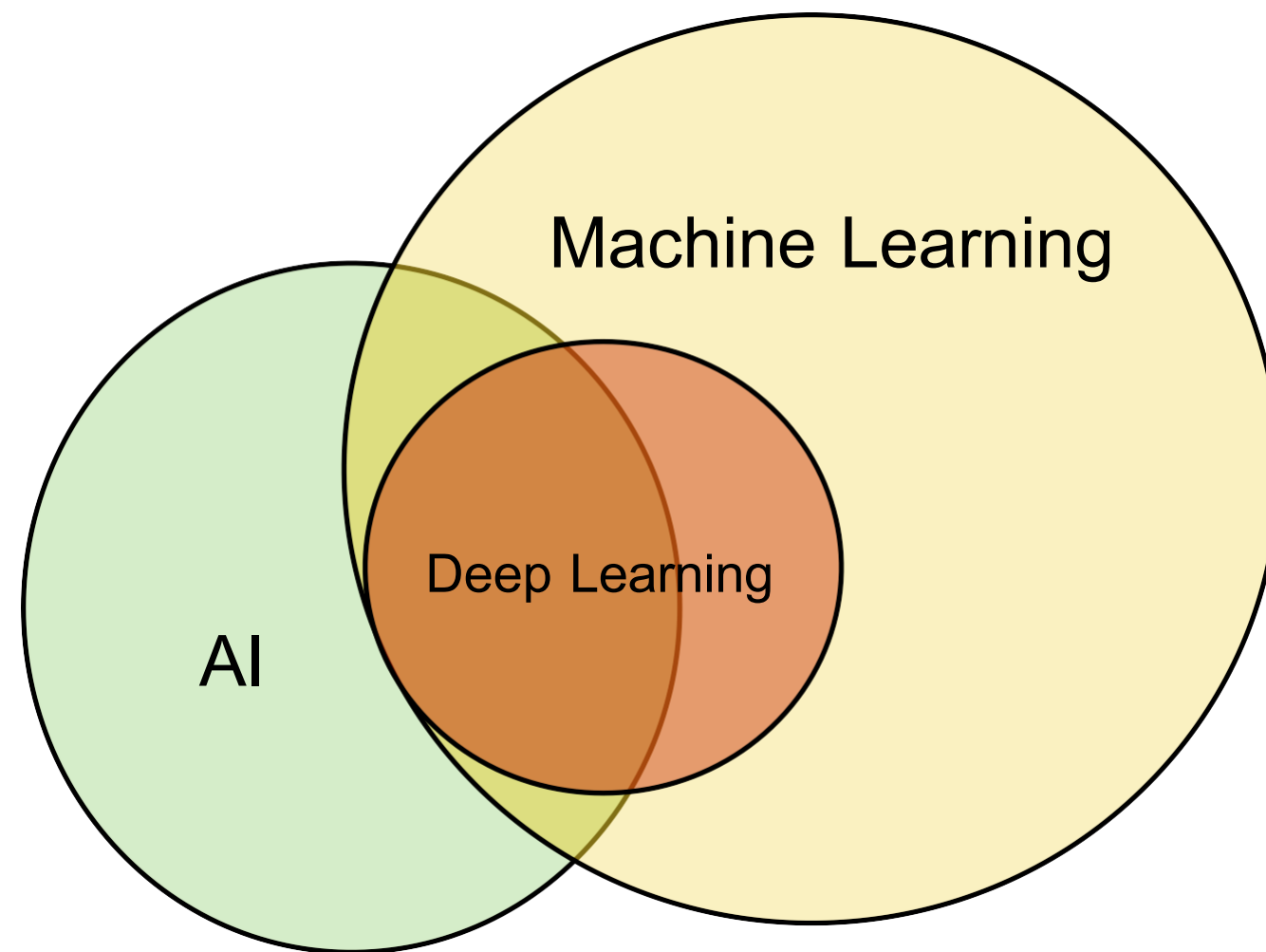


Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed
– Arthur Samuel (1959)

Machine Learning



The Connection Between Fields



Different Types Of AI

Artificial Intelligence (AI):

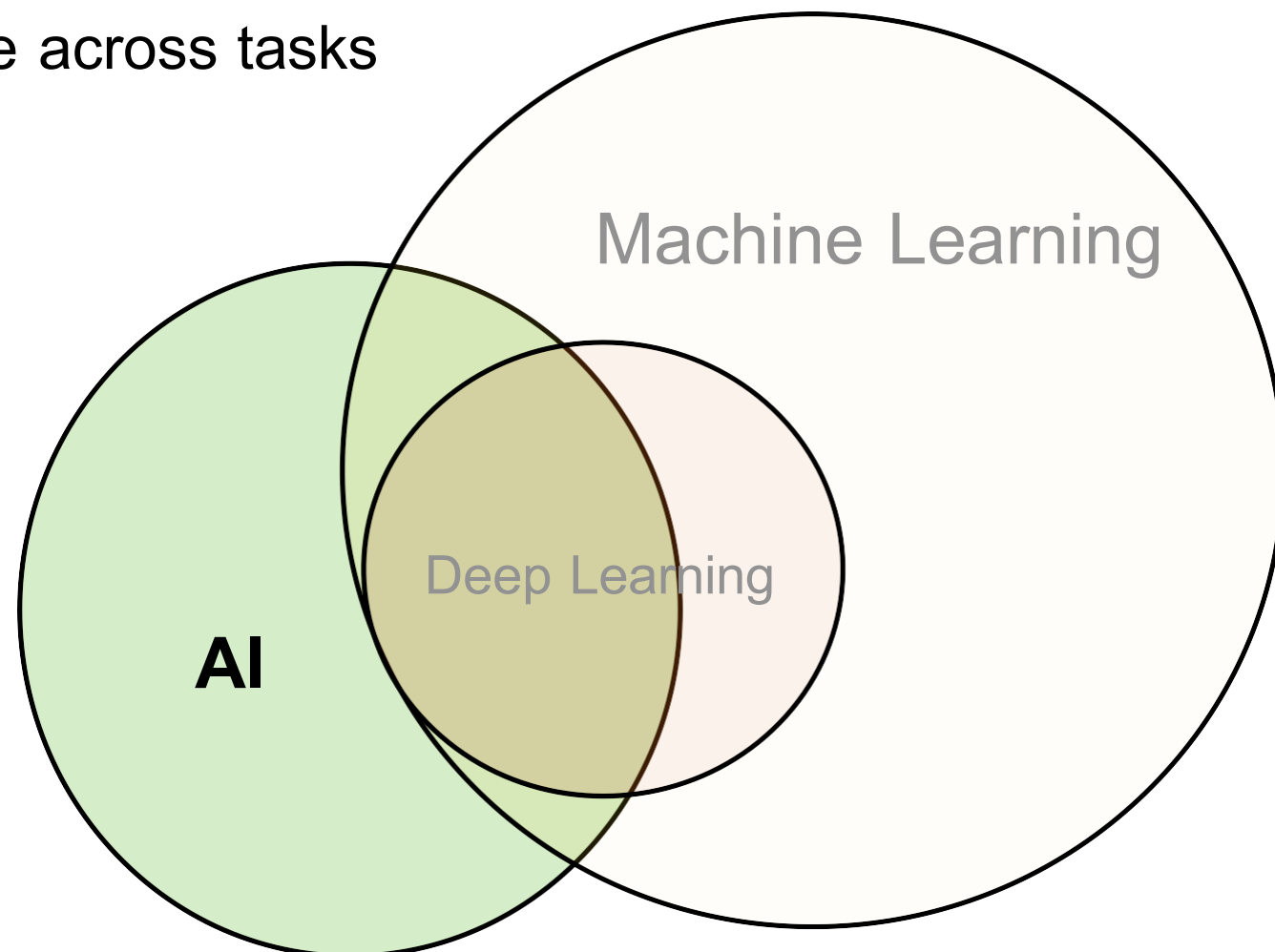
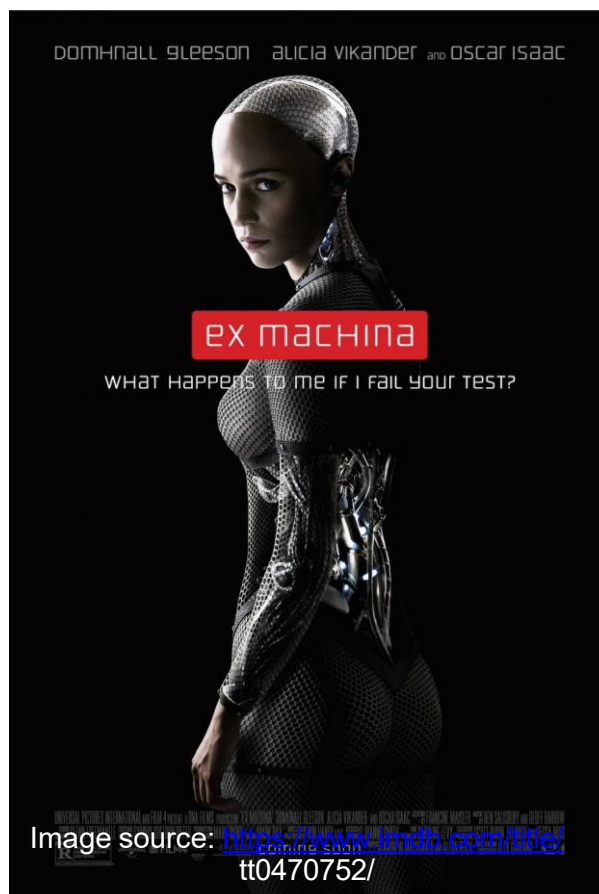
orig. subfield of computer science, solving tasks humans are good at (natural language, speech, image recognition, ...)

Narrow AI:

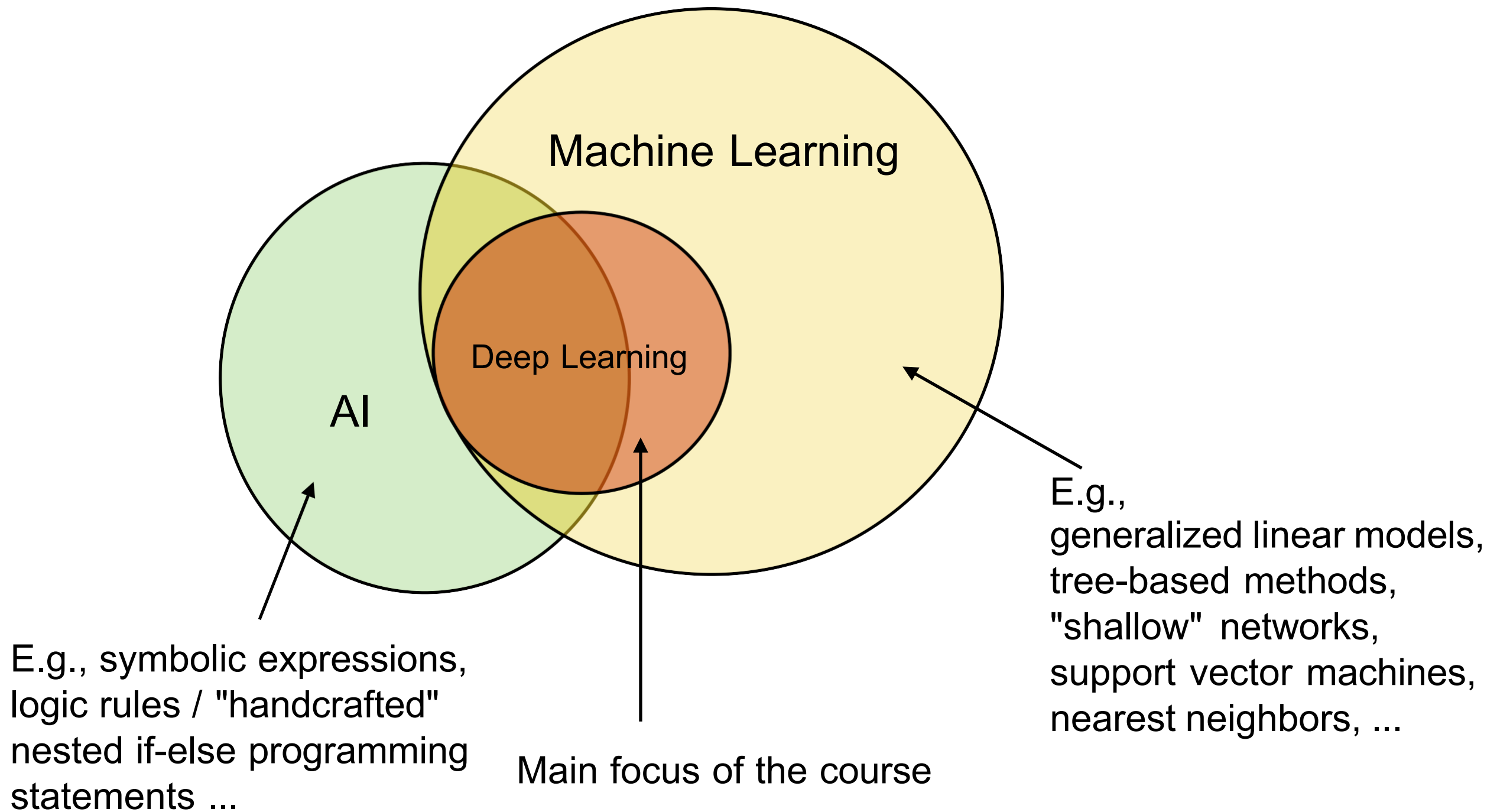
solving a particular task (playing a game, driving a car, ...)

Artificial General Intelligence (AGI):

multi-purpose AI mimicking human intelligence across tasks



What This Course Is About



Not all AI Systems involve Machine Learning

Deep Blue used custom VLSI chips to execute the **alpha-beta search** algorithm in parallel, an example of GOFAI (Good Old-Fashioned Artificial Intelligence).

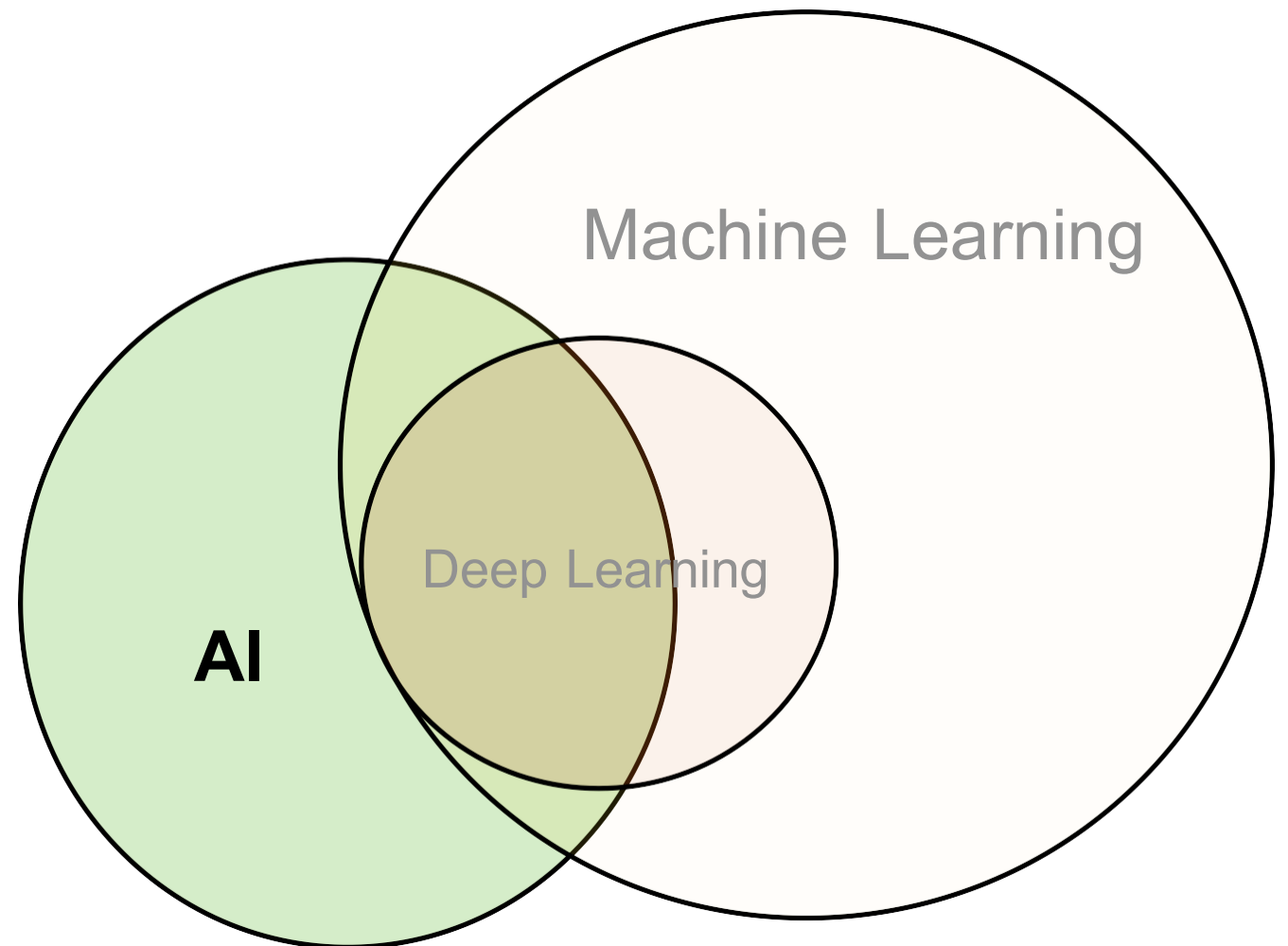
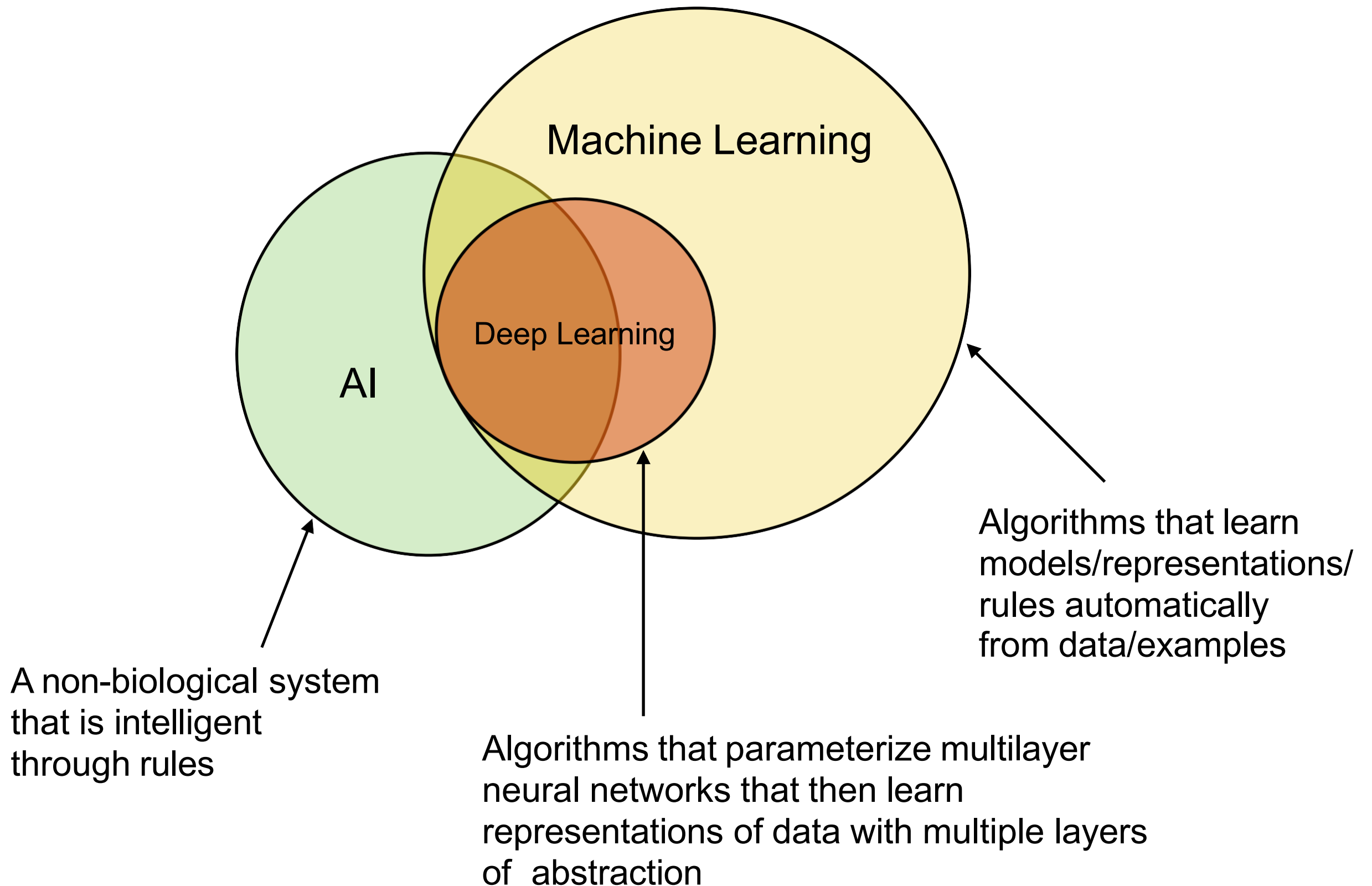


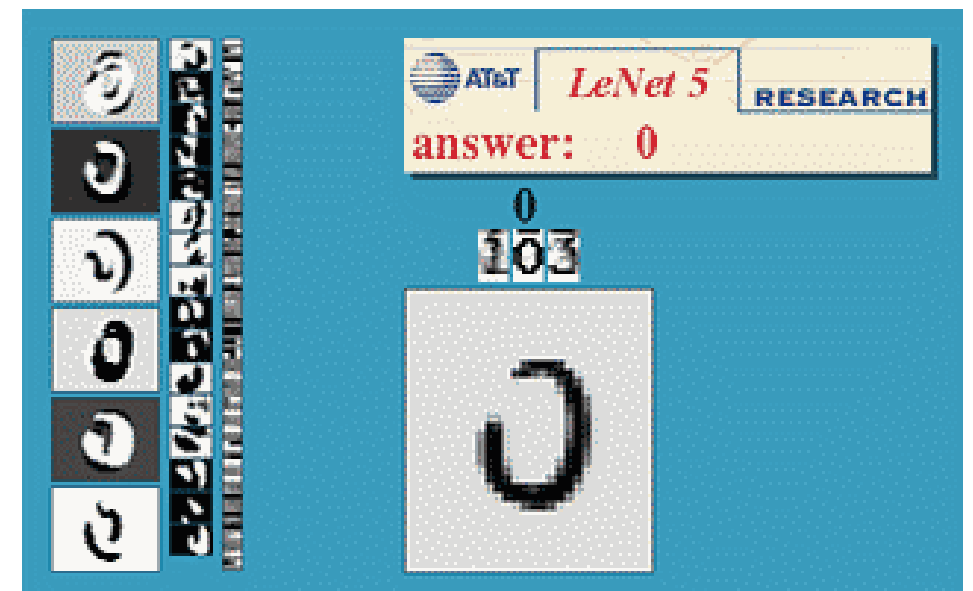
Image Source: <https://mashable.com/2016/02/10/kasparov-deep-blue>

Examples From The Three Related "Areas"



Some Applications Of Machine Learning/Deep Learning

- Email spam detection
- Fingerprint / face detection & matching (e.g., phones)
- Web search (e.g., DuckDuckGo, Bing, Google)
- Sports predictions
- Post office (e.g., sorting letters by zip codes)
- ATMs (e.g., reading checks)
- Credit card fraud
- Stock predictions



Source: <http://yann.lecun.com/exdb/lenet/>

Some Applications Of Machine Learning/Deep Learning

- Smart assistants (Apple Siri, Amazon Alexa, ...)
- Product recommendations (e.g., Netflix, Amazon)
- Self-driving cars (e.g., Uber, Tesla)
- Language translation (Google translate)
- Sentiment analysis
- Drug design
- Medical diagnoses
- ...



Source: <https://techcrunch.com/2017/11/07/waymo-now-testing-its-self-driving-cars-on-public-roads-with-no-one-at-the-wheel/>

The 3 Broad Categories of ML

(This also applies to DL)

1. Course overview
2. What is machine learning?
- 3. The broad categories of ML**
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The 3 Broad Categories Of ML (And DL)

Supervised Learning

- Labeled data
- Direct feedback
- Predict outcome/future

Unsupervised Learning

- No labels/targets
- No feedback
- Find hidden structure in data

Reinforcement Learning

- Decision process
- Reward system
- Learn series of actions

Source: Raschka and Mirjalily (2019). *Python Machine Learning, 3rd Edition*

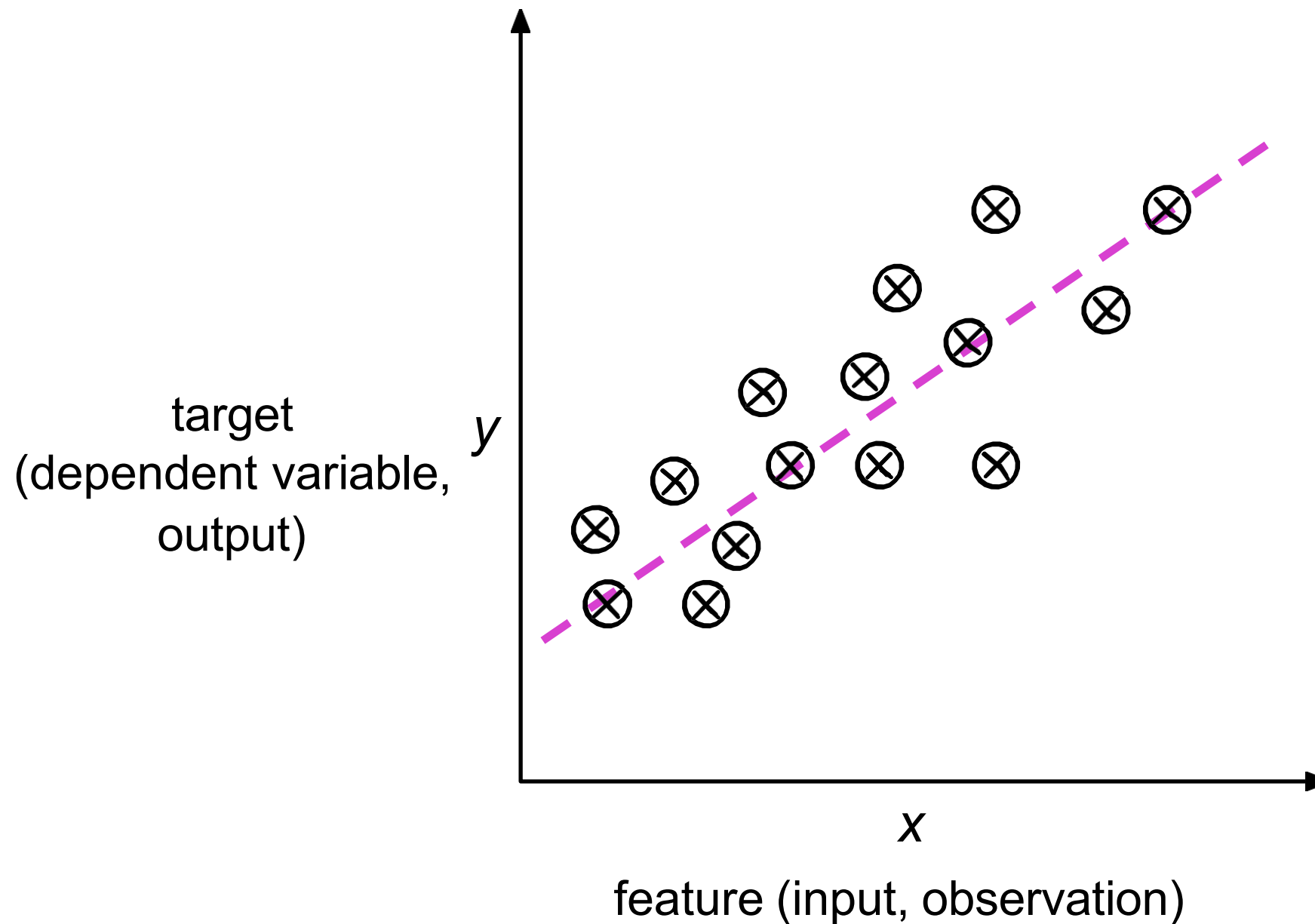
Supervised Learning Is The Largest Subcategory

Supervised Learning

- Labeled data
- Direct feedback
- Predict outcome/future

Source: Raschka and Mirjalily (2019). *Python Machine Learning, 3rd Edition*

Supervised Learning 1: Regression

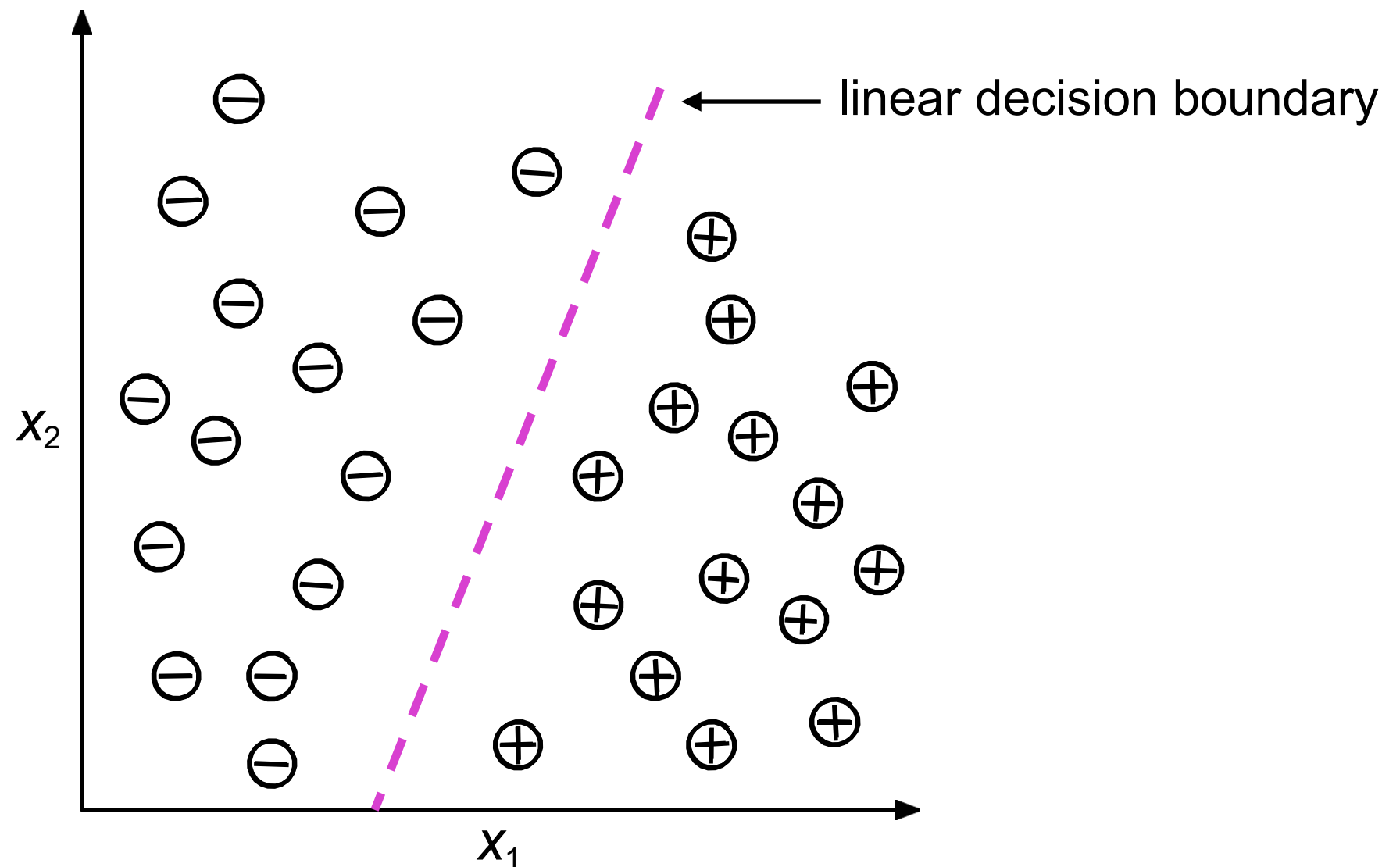


Source: Raschka and Mirjalili (2019). *Python Machine Learning, 3rd Edition*

Supervised Learning 2: Classification

Binary classification example with two *features* ("independent" variables, predictors)

What are the
class labels (y's)?



Source: Raschka and Mirjalily (2019). *Python Machine Learning, 3rd Edition*

Supervised Learning 3: Ordinal regression

- Ordinal regression also called *ordinal classification* or *ranking* (although ranking is a bit different)

Order dependence like in metric regression,
but no metric distance

discrete values like in classification,
but order dependence

$$r_K \succ r_{K-1} \succ \dots \succ r_1$$

E.g., movie ratings: *great* $\succ$ *good* $\succ$ *okay* $\succ$ *for genre fans* $\succ$ *bad*

Supervised Learning 3: Ordinal regression

- **Ranking:** Predict Correct order
(0 loss if order is correct, e.g., rank a collection of movies by "goodness")



- **Ordinal regression:** Predict correct (ordered) label
(E.g., age of a person in years; here, regard aging as a non-stationary process)



18



29



41

Excerpt from the UTKFace dataset
<https://susanqq.github.io/UTKFace/>

Supervised Learning 3: Ordinal regression

- **Ranking:** Predict correct order
(0 loss if order is correct, e.g., rank a collection of movies by "goodness")



- **Ordinal regression:** Predict correct (ordered) label
(E.g., age of a person in years; here, regard aging as a non-stationary process)

Excerpt from the UTKFace dataset
<https://susanqq.github.io/UTKFace/>



18



29



41

The 2nd Subcategory Of ML (And DL)

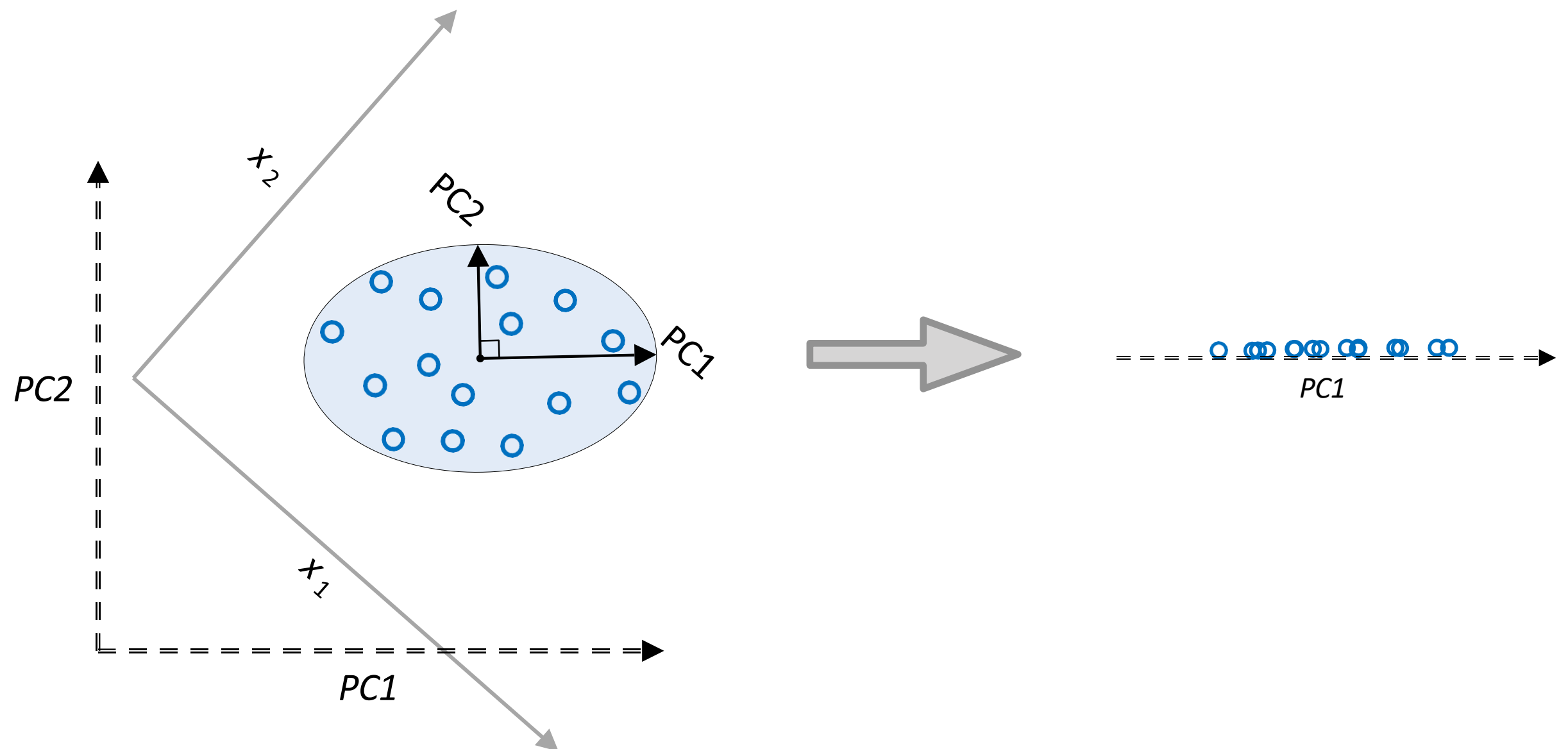
Unsupervised Learning

- No labels/targets
- No feedback
- Find hidden structure in data

Source: Raschka and Mirjalily (2019). *Python Machine Learning, 3rd Edition*

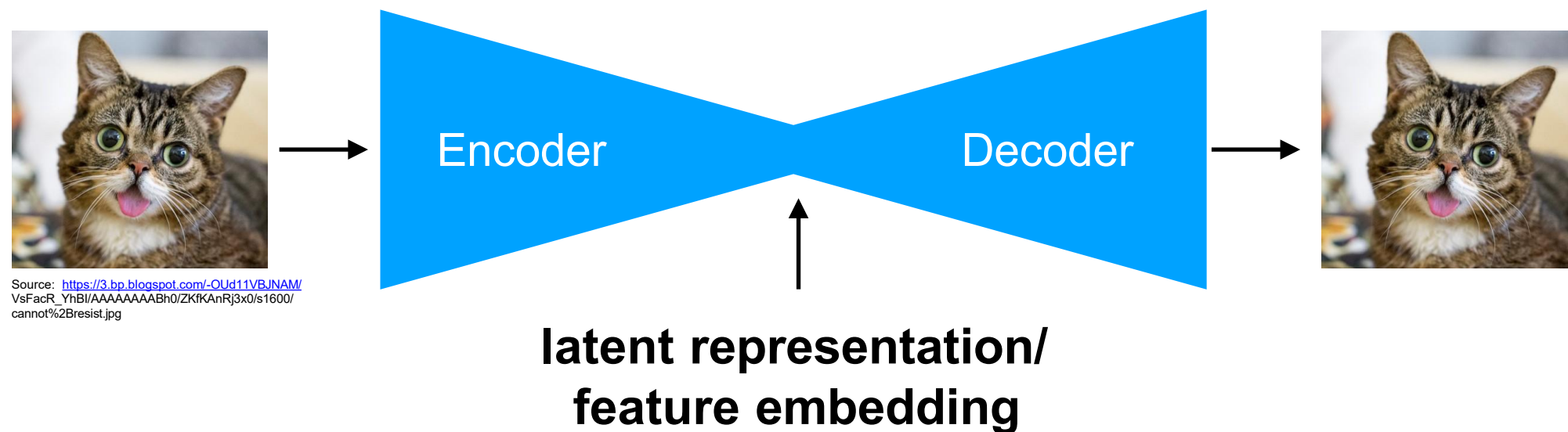
Unsupervised Learning 1: Representation Learning/Dimensionality Reduction

E.g., Principal Component Analysis (PCA)



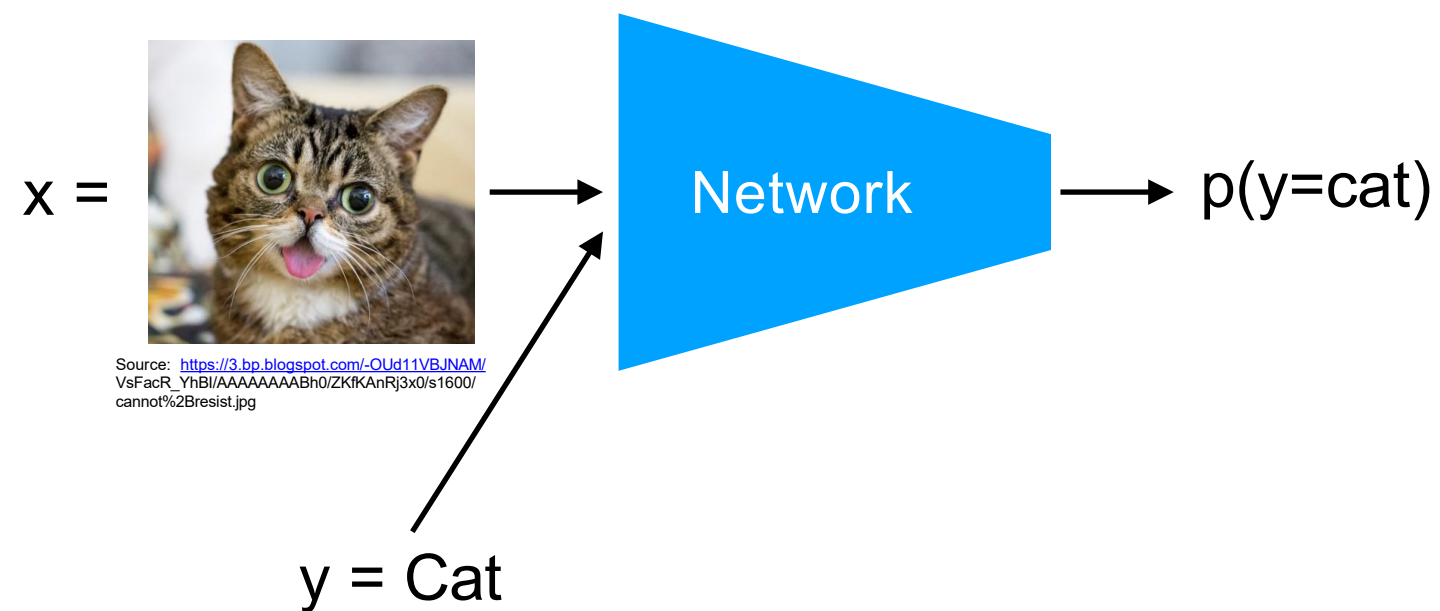
Unsupervised Learning 1: Representation Learning/Dimensionality Reduction

E.g., Autoencoders



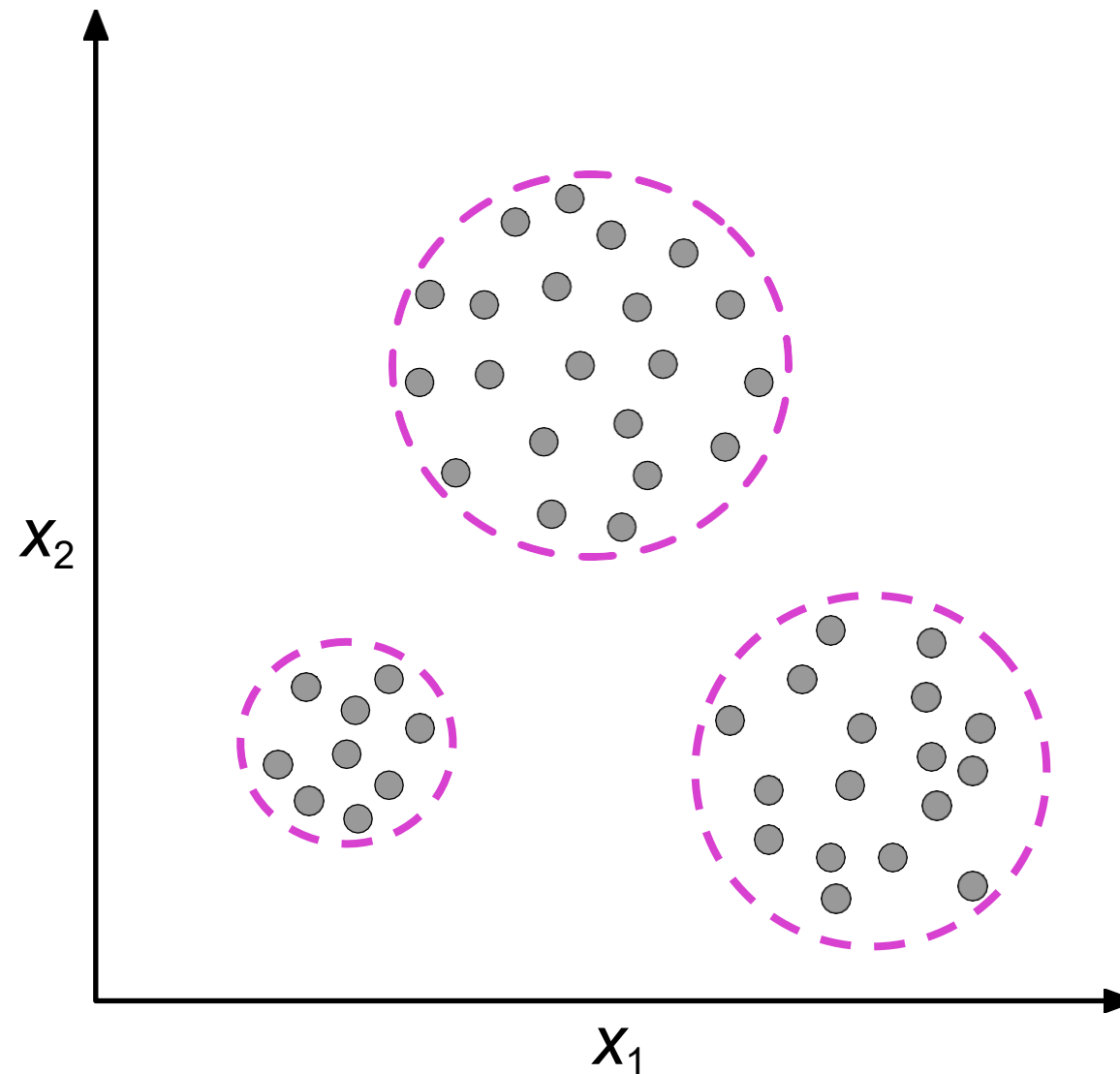
(covered later in this course)

Reminder: Classification works like this



Unsupervised Learning 2: Clustering

Assigning group memberships to unlabelled examples (instances, data points)



Source: Raschka and Mirjalily (2019). *Python Machine Learning, 3rd Edition*

Reinforcement Learning: The third subcategory of ML (and DL)

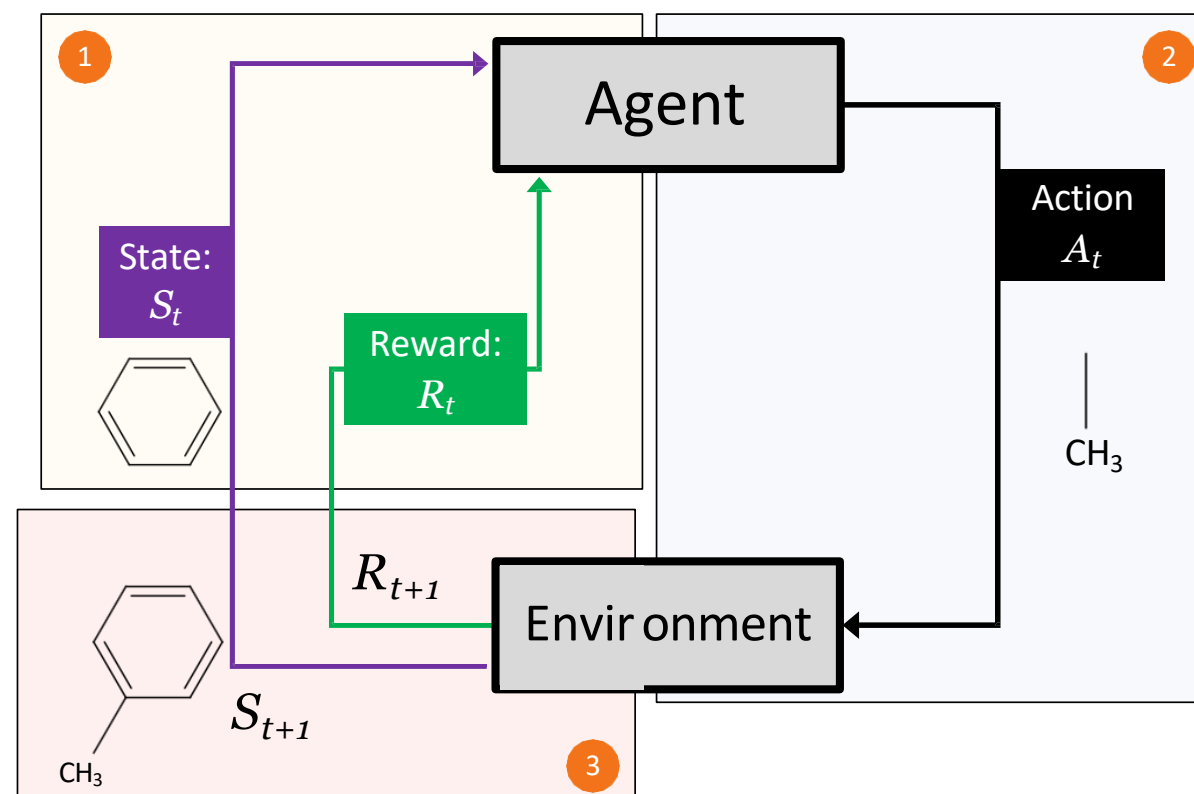


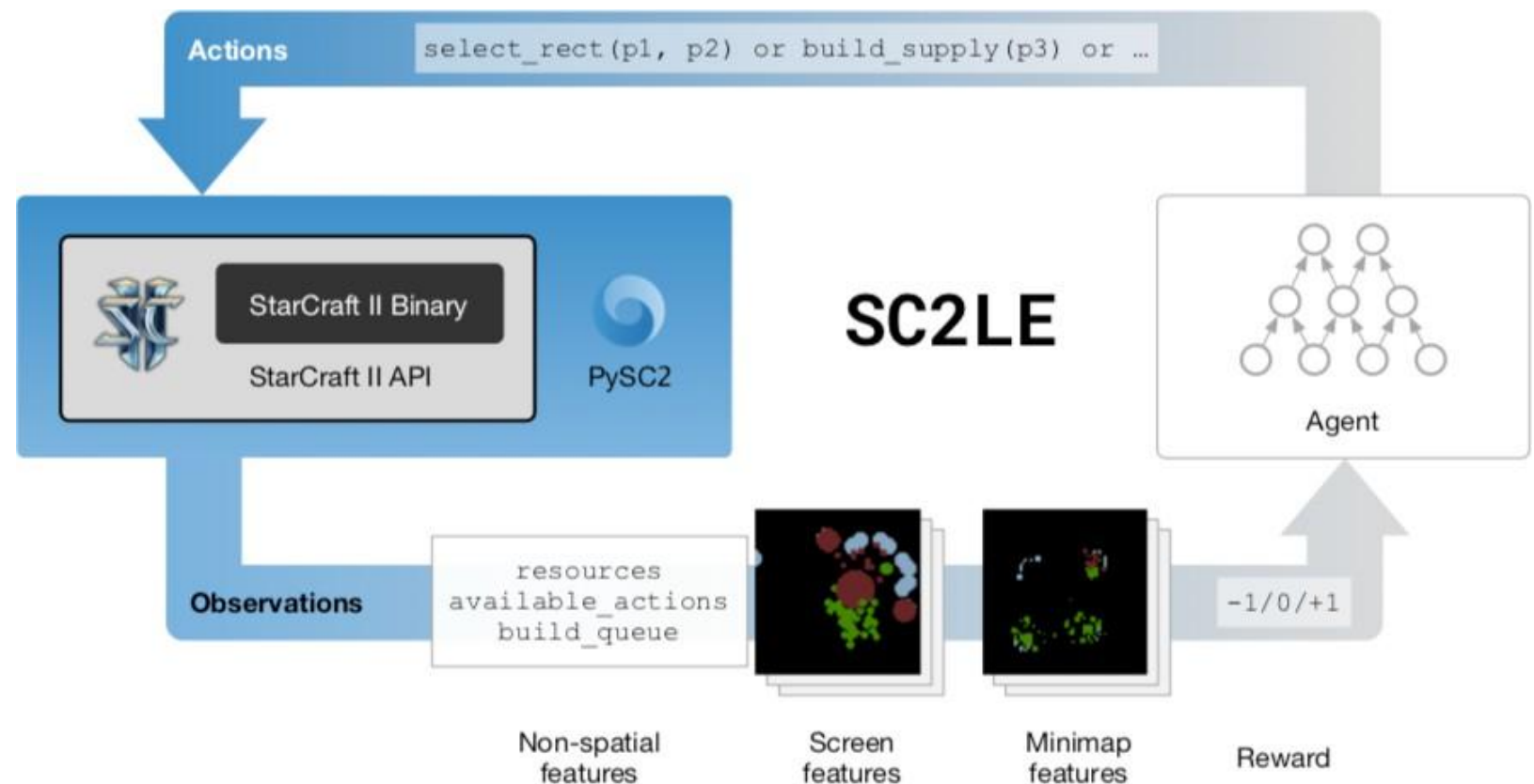
Figure 5: Representation of the basic reinforcement learning paradigm with a simple molecular example. (1) Given a benzene ring (state S_t at iteration t) and some reward value R_t at iteration t , (2) the agent selects an action A_t that adds a methyl group to the benzene ring. (3) The environment considers this information for producing the next state (S_{t+1}) and reward (R_{t+1}). This cycle repeats until the episode is terminated.

Source: Sebastian Raschka and Benjamin Kaufman (2020)

Machine learning and AI-based approaches for bioactive ligand discovery and GPCR-ligand recognition

Reinforcement Learning:

The third subcategory of ML (and DL)



Vinyals, Oriol, Timo Ewalds, Sergey Bartunov, Petko Georgiev, Alexander Sasha Vezhnevets, Michelle Yeo, Alireza Makhzani et al. "Starcraft II: A new challenge for reinforcement learning." *arXiv preprint arXiv:1708.04782* (2017).

Semi-Supervised Learning

- mix between supervised and unsupervised learning
- some training examples contain outputs, but some do not
- use the labeled training subset to label the unlabeled portion of the training set, which we then also utilize for model training

Semi-Supervised Learning

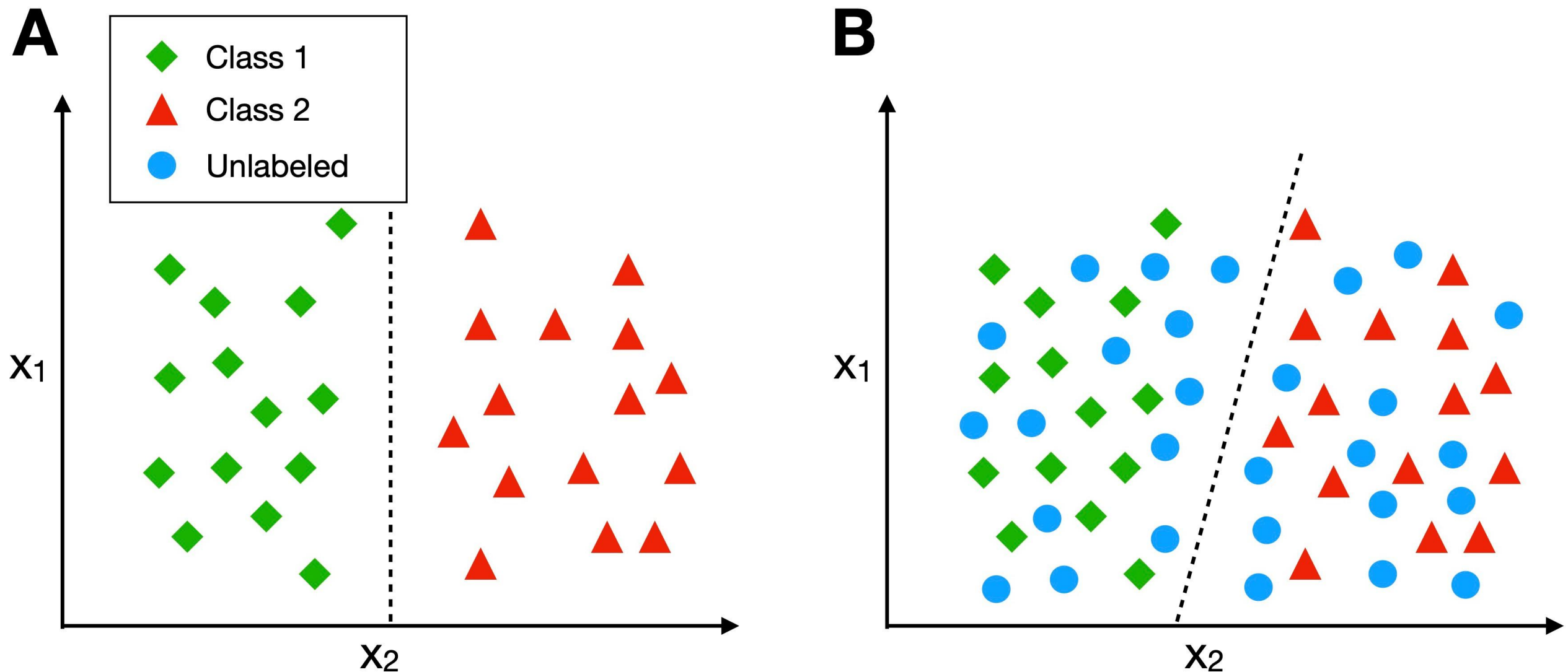


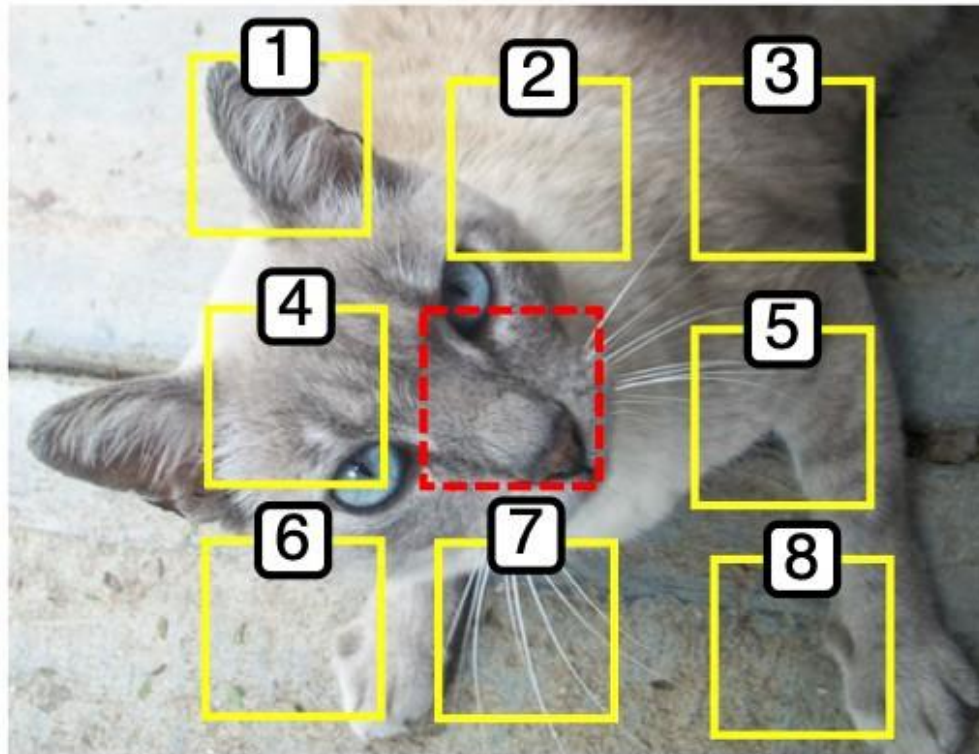
Illustration of semi-supervised learning incorporating unlabeled examples. (A) A decision boundary derived from the labeled training examples only. (B) A decision boundary based on both labeled and unlabeled examples.

Self-Supervised Learning

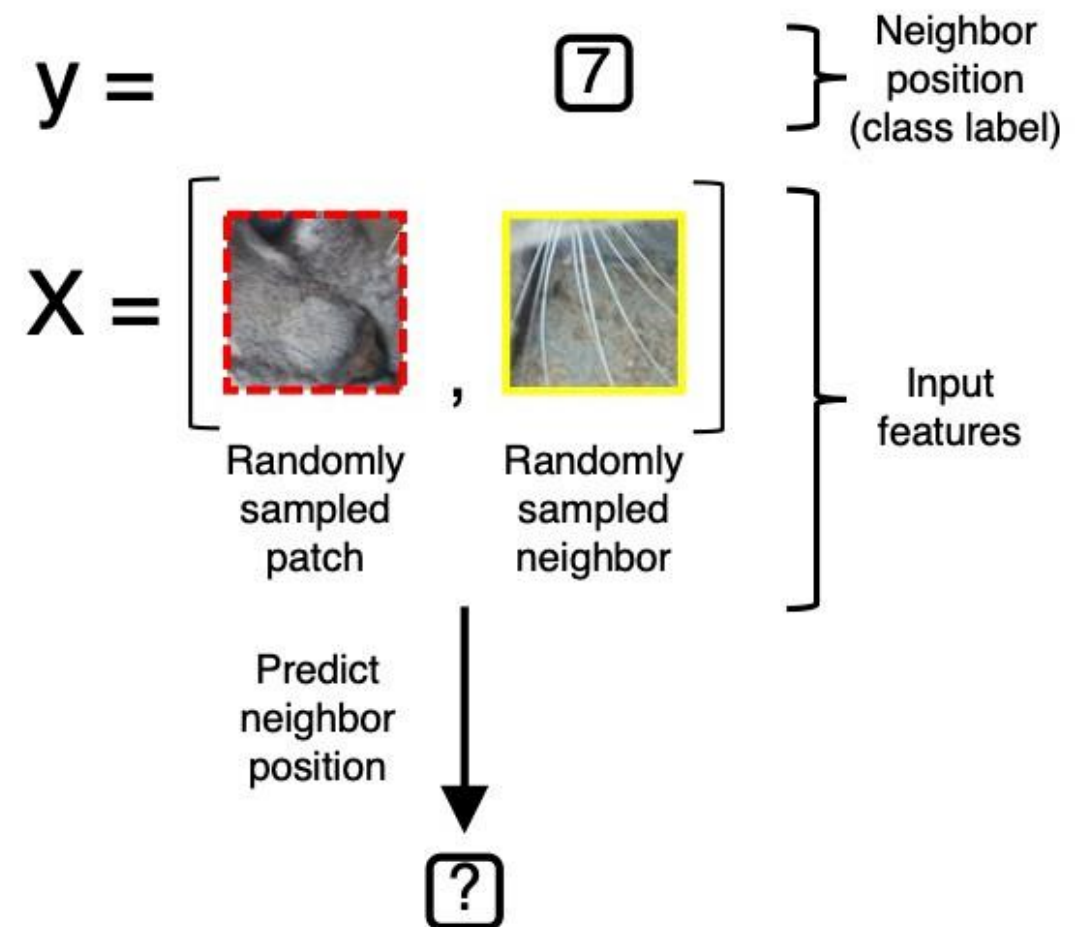
- A recent development and promising research trend in deep learning
- particularly useful if pre-trained models for transfer learning are not available for the target domain
- a process of deriving and utilizing label information directly from the data itself rather than having humans annotating it

Self-Supervised Learning

A



B

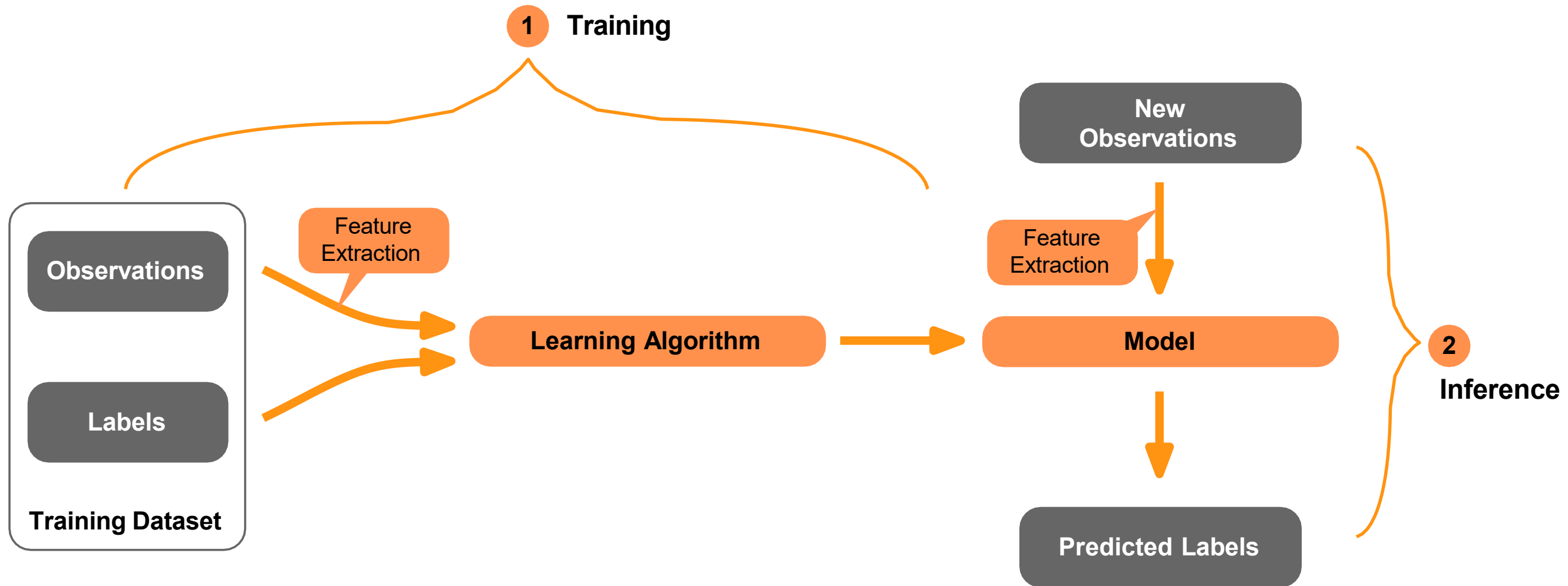


Self-supervised learning via context prediction. (A) A random patch is sampled (red square) along with 9 neighboring patches. (B) Given the random patch and a random neighbor patch, the task is to predict the position of the neighboring patch relative to the center patch (red square).

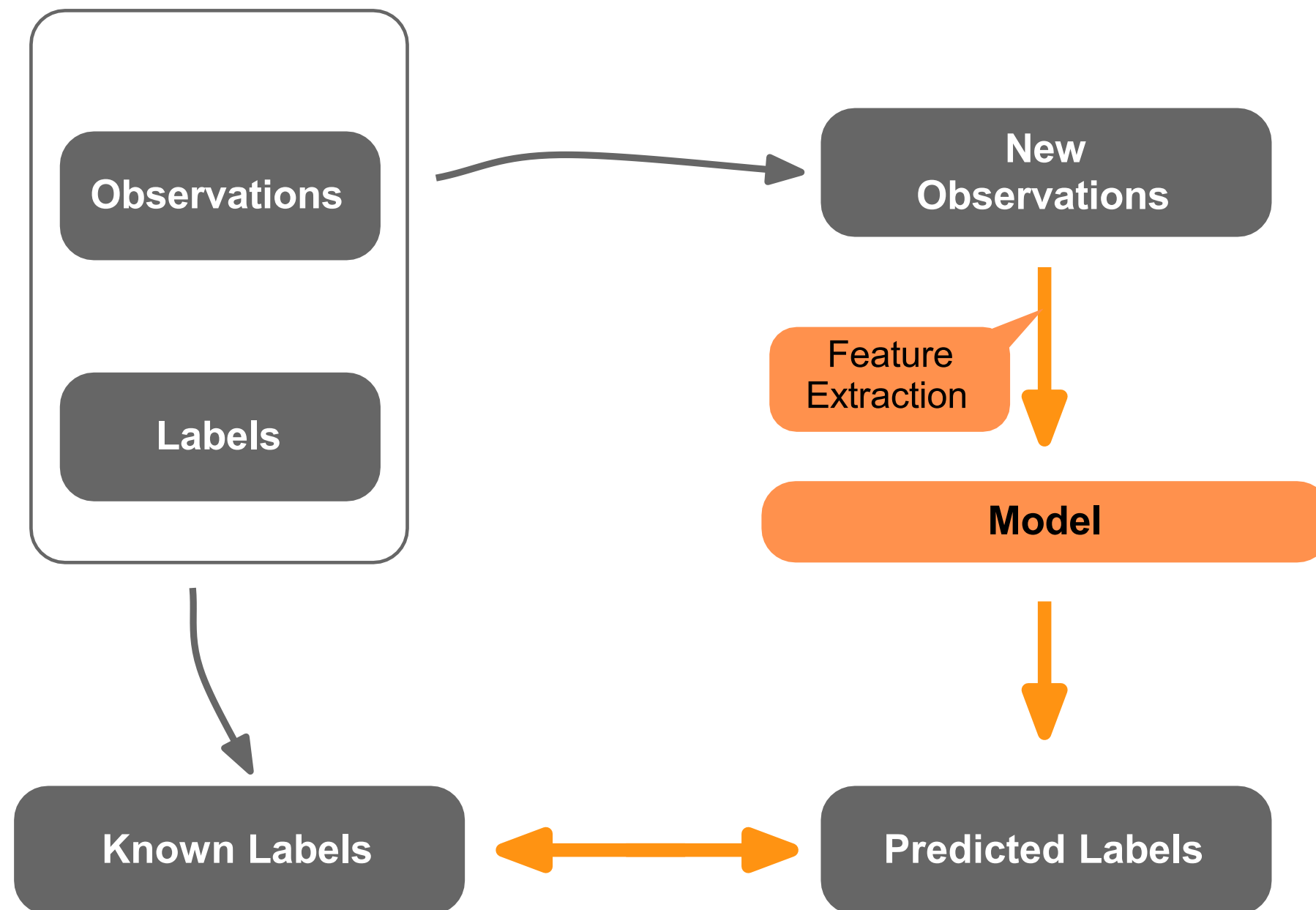
The Supervised Learning Workflow

1. Course overview
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Supervised Learning Workflow



Using a test dataset to evaluate the performance of a predictive model



Structured vs Unstructured Data

A

Feature vector of the 1st training example

Class label

Index	Sepal length	Sepal width	Petal length	Petal width	Species
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
...	...	...	...	...	...
150	5.9	3	5.1	1.8	Iris-virginica

B



Machine Learning vs Deep Learning

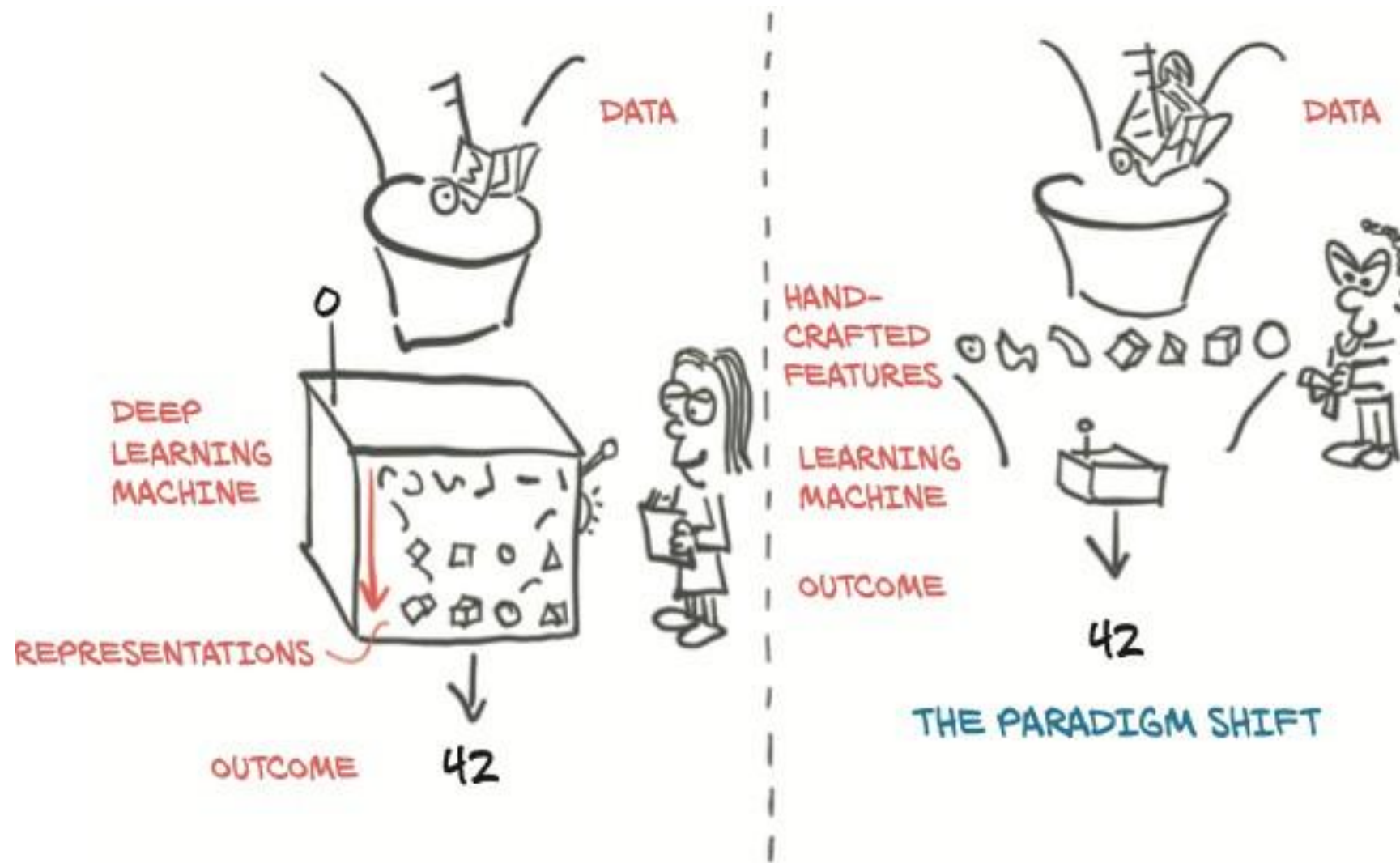


Image source: Stevens et al., *Deep Learning with PyTorch*. Manning, 2020

Machine Learning vs Deep Learning

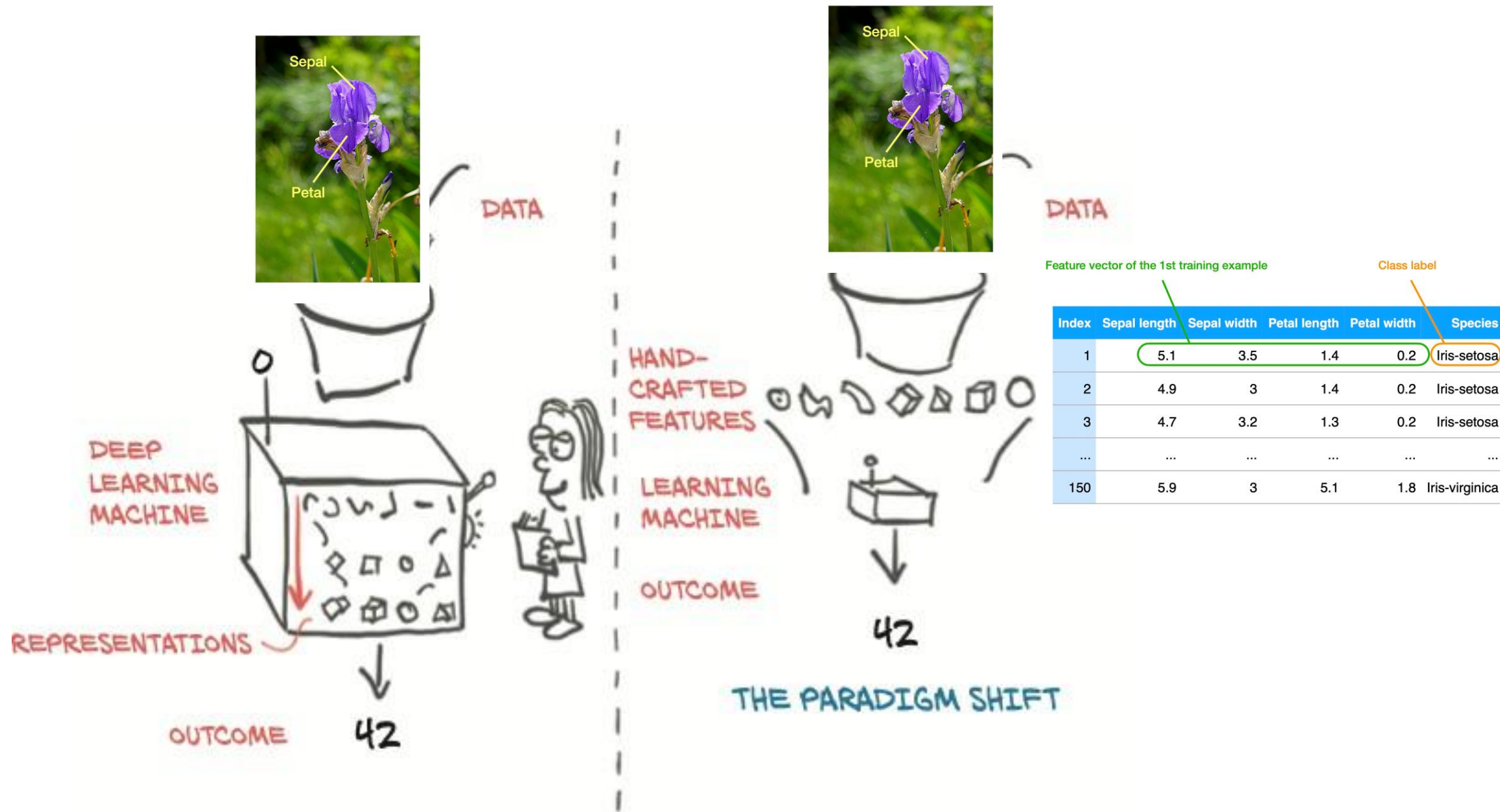


Image source: Stevens et al., *Deep Learning with PyTorch*. Manning, 2020

Machine Learning

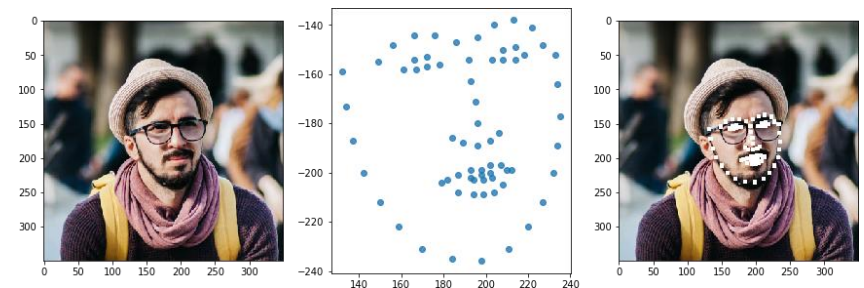
Terminology and Notation

(Again, this also applies to DL)

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Machine Learning Jargon 1/2

- ***supervised learning:***
learn function to map input x (features) to output y (targets)
- ***structured data:***
databases, spreadsheets/csv files
- ***unstructured data:***
features like image pixels, audio signals, text sentences
(before DL, extensive feature engineering was required)



Source: http://rasbt.github.io/mlxtend/user_guide/image/extract_face_landmarks/

Supervised Learning (More Formal Notation)

"training examples"

Training set: $\mathcal{D} = \{\langle \mathbf{x}^{[i]}, y^{[i]} \rangle, i = 1, \dots, n\},$

Unknown function: $f(\mathbf{x}) = y$

Hypothesis: $h(\mathbf{x}) = \hat{y}$

sometimes t or o

Classification

Regression

$$h : \mathbb{R}^m \rightarrow \mathcal{Y}, \quad \mathcal{Y} = \{1, \dots, k\}$$

$$h : \mathbb{R}^m \rightarrow \mathbb{R}$$

Data Representation

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

Feature vector

Data Representation

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

Feature vector

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix}$$

Design Matrix

$$\mathbf{X} = \begin{bmatrix} x_1^{[1]} & x_2^{[1]} & \dots & x_m^{[1]} \\ x_1^{[2]} & x_2^{[2]} & \dots & x_m^{[2]} \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{[n]} & x_2^{[n]} & \dots & x_m^{[n]} \end{bmatrix}$$

Design Matrix

Data Representation (structured data)

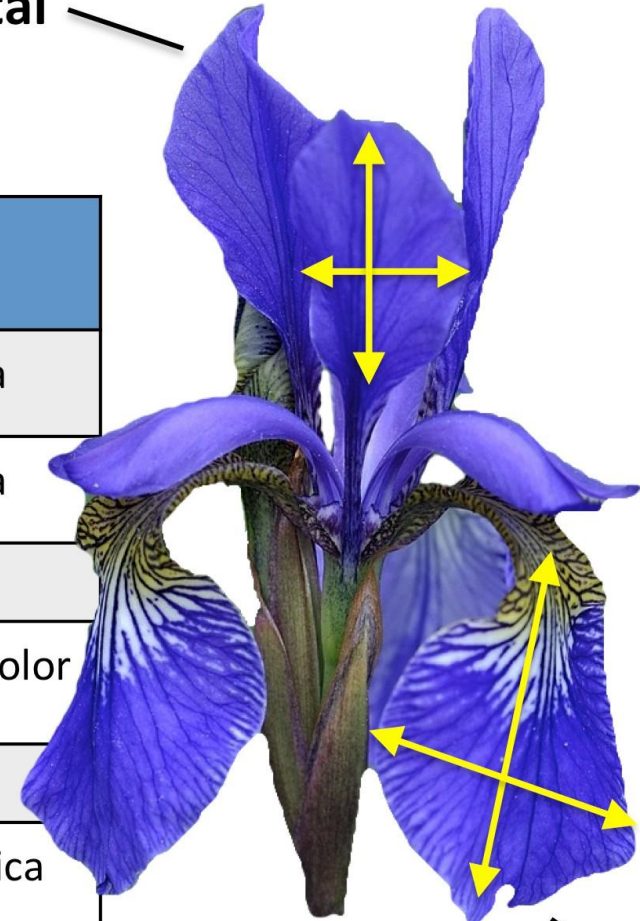
$m =$ _____

$n =$ _____



	Sepal length	Sepal width	Petal length	Petal width	
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Petal



Sepal



Data Representation (unstructured data; images)

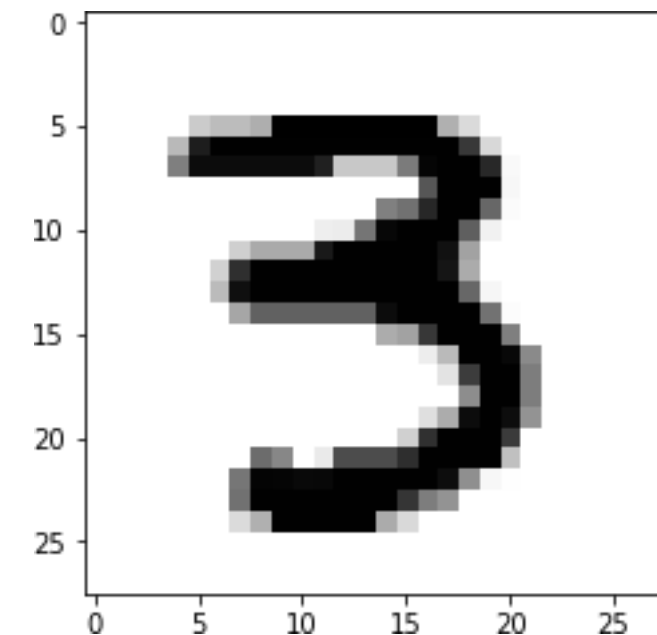
Convolutional Neural Networks

Image batch dimensions: `torch.Size([128, 1, 28, 28])` ← "NCHW" representation (more on that later)

Image label dimensions: `torch.Size([128])`

```
print(images[0].size())  
  
torch.Size([1, 28, 28])
```

```
images[0]  
  
tensor([[[[0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000],  
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000],  
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000],  
         [0.0000, 0.0000, 0.0000, 0.0000, 0.5020, 0.9529, 0.9529, 0.9529,  
          0.9529, 0.9529, 0.9529, 0.8706, 0.2157, 0.2157, 0.2157, 0.5176,  
          0.9804, 0.9922, 0.9922, 0.8392, 0.0235, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000],  
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.6627, 0.9922, 0.9922, 0.9922, 0.0314, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000],  
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.4980, 0.5529,  
          0.8471, 0.9922, 0.9922, 0.5961, 0.0157, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0000],  
         [0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000,  
          0.0000, 0.0000, 0.0000, 0.0667, 0.0745, 0.5412, 0.9725, 0.9922,  
          0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000]]]])
```



Machine Learning Jargon 2/2

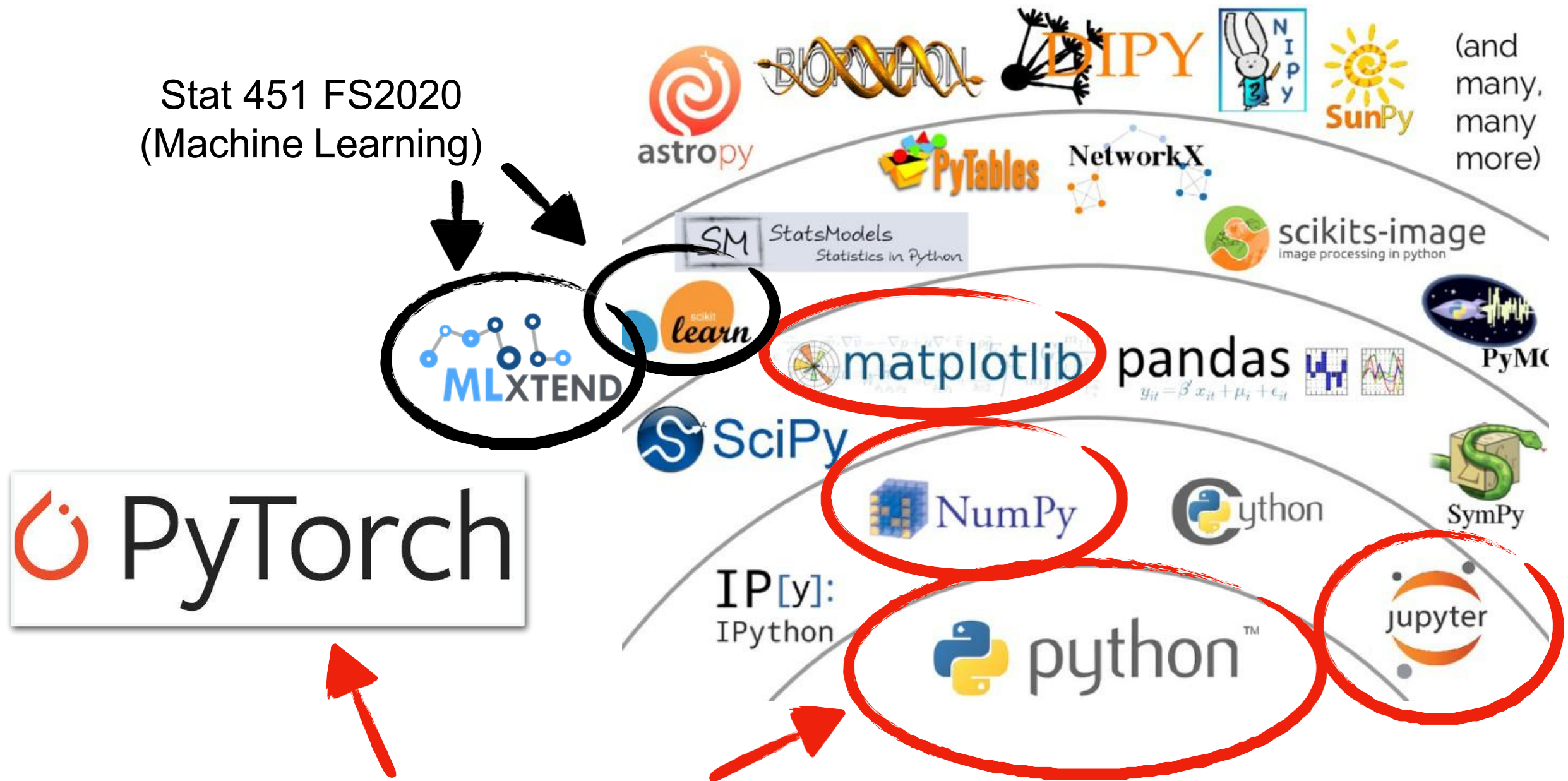
- **Training a model** = fitting a model = parameterizing a model = learning from data
- **Training example**, synonymous to training record, training instance, training sample (in some contexts, sample refers to a collection of training examples)
- **Feature**, synonymous to observation, predictor, variable, independent variable, input, attribute, covariate
- **Target**, synonymous to outcome, ground truth, output, response variable, dependent variable, (class) label (in classification)
- **Output / Prediction**, use this to distinguish from targets; here, means output from the model

The Practical Aspects: Our Tools!

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Main Scientific Python Libraries

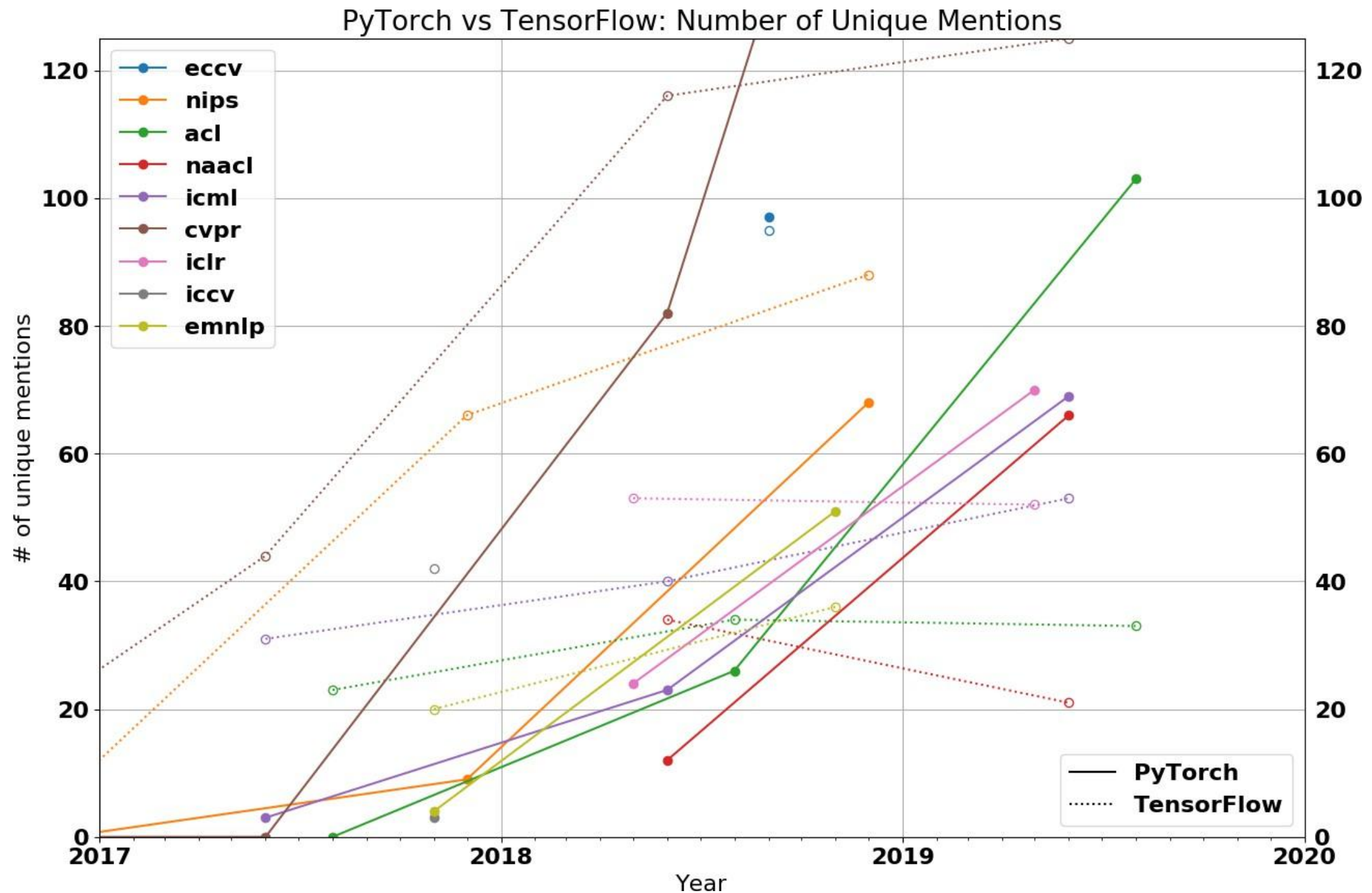
Stat 451 FS2020
(Machine Learning)



Main tools for this course

Image by Jake VanderPlas. Source:
<https://speakerdeck.com/jakevdp/the-state-of-the-stack-scipy-2015-keynote?slide=8>

"The State of Machine Learning Frameworks in 2019"



Source:

<https://thegradient.pub/state-of-ml-frameworks-2019-pytorch-dominates-research-tensorflow-dominates-industry/>

"The State of Machine Learning Frameworks in 2019"

CONFERENCE	PT 2018	PT 2019	PT GROWTH	TF 2018	TF 2019	TF GROWTH
CVPR	82	280	240%	116	125	7.7%
NAACL	12	66	450%	34	21	-38.2%
ACL	26	103	296%	34	33	-2.9%
ICLR	24	70	192%	54	53	-1.9%
ICML	23	69	200%	40	53	32.5%

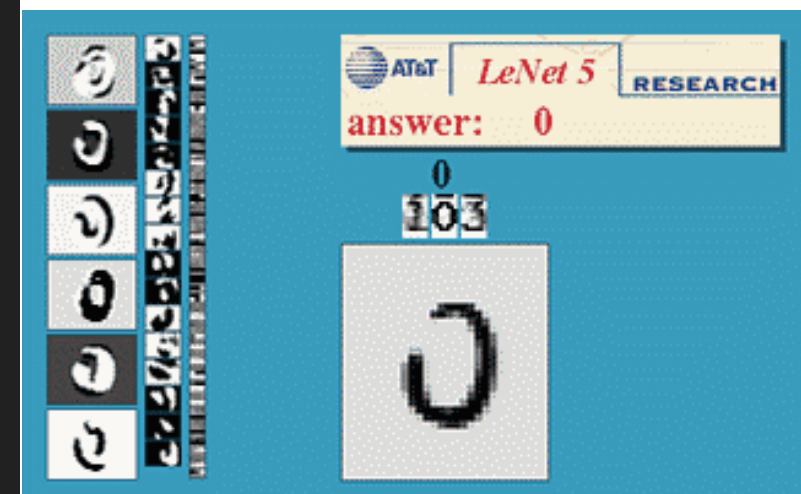
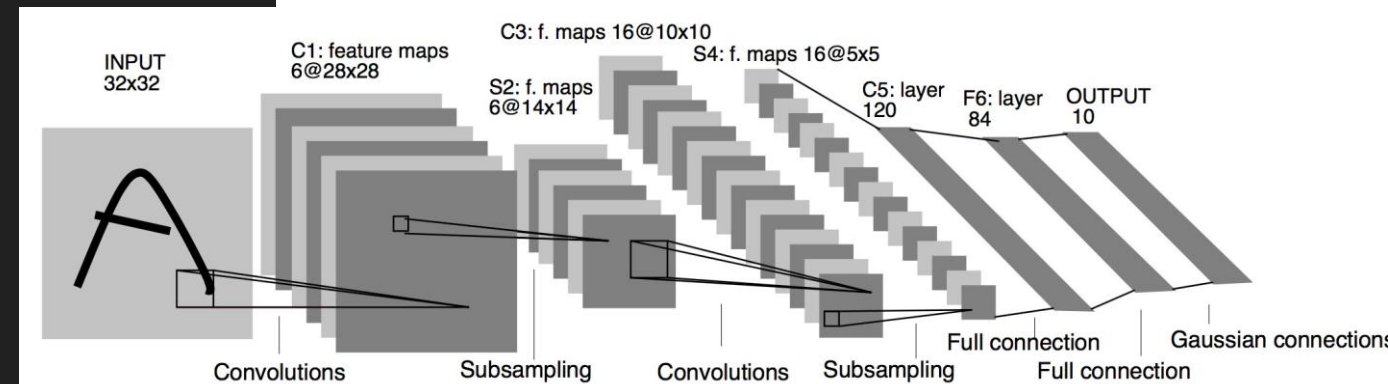
In 2018, PyTorch was a minority. Now, it is an overwhelming majority, with 69% of CVPR using PyTorch, 75+% of both NAACL and ACL, and 50+% of ICLR and ICML. While PyTorch's dominance is strongest at vision and language conferences (outnumbering TensorFlow by 2:1 and 3:1 respectively), PyTorch is also more popular than TensorFlow at general machine learning conferences like ICLR and ICML.

Source:
<https://thegradient.pub/state-of-ml-frameworks-2019-pytorch-dominates-research-tensorflow-dominates-industry/>

```

34 #####
35 # ## MODEL
36 #####
37
38 class LeNet5(torch.nn.Module):
39
40     def __init__(self, num_classes):
41         super().__init__()
42
43         self.features = torch.nn.Sequential(
44             torch.nn.Conv2d(1, 6, kernel_size=5),
45             torch.nn.Tanh(),
46             torch.nn.MaxPool2d(kernel_size=2),
47             torch.nn.Conv2d(6, 16, kernel_size=5),
48             torch.nn.Tanh(),
49             torch.nn.MaxPool2d(kernel_size=2)
50         )
51
52         self.classifier = torch.nn.Sequential(
53             torch.nn.Linear(16*5*5, 120),
54             torch.nn.Tanh(),
55             torch.nn.Linear(120, 84),
56             torch.nn.Tanh(),
57             torch.nn.Linear(84, num_classes),
58         )
59
60     def forward(self, x):
61         x = self.features(x)
62         x = torch.flatten(x, 1)
63         logits = self.classifier(x)
64         probas = torch.nn.functional.softmax(logits, dim=1)
65         return logits, probas
66
67

```



Source: <http://yann.lecun.com/exdb/lenet/>

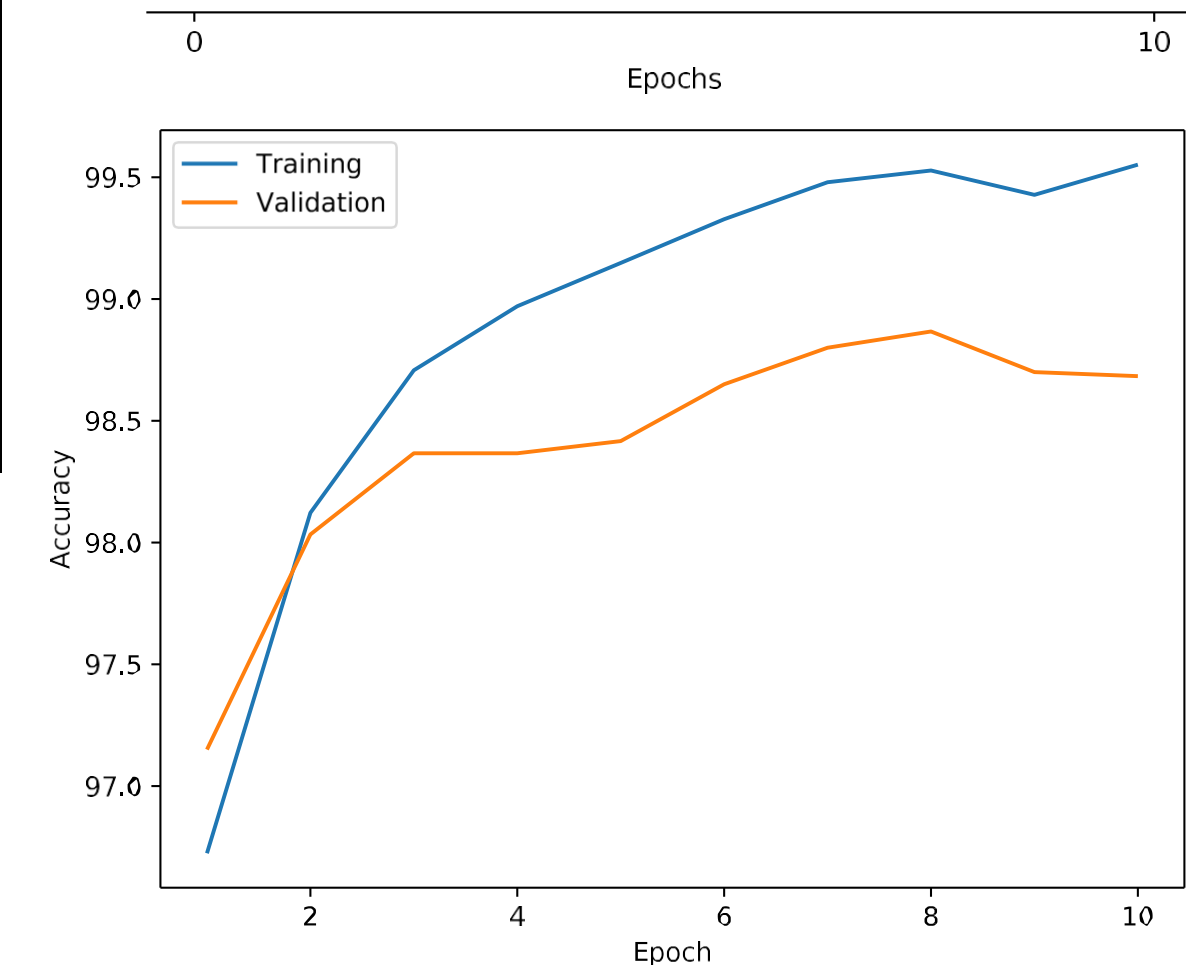
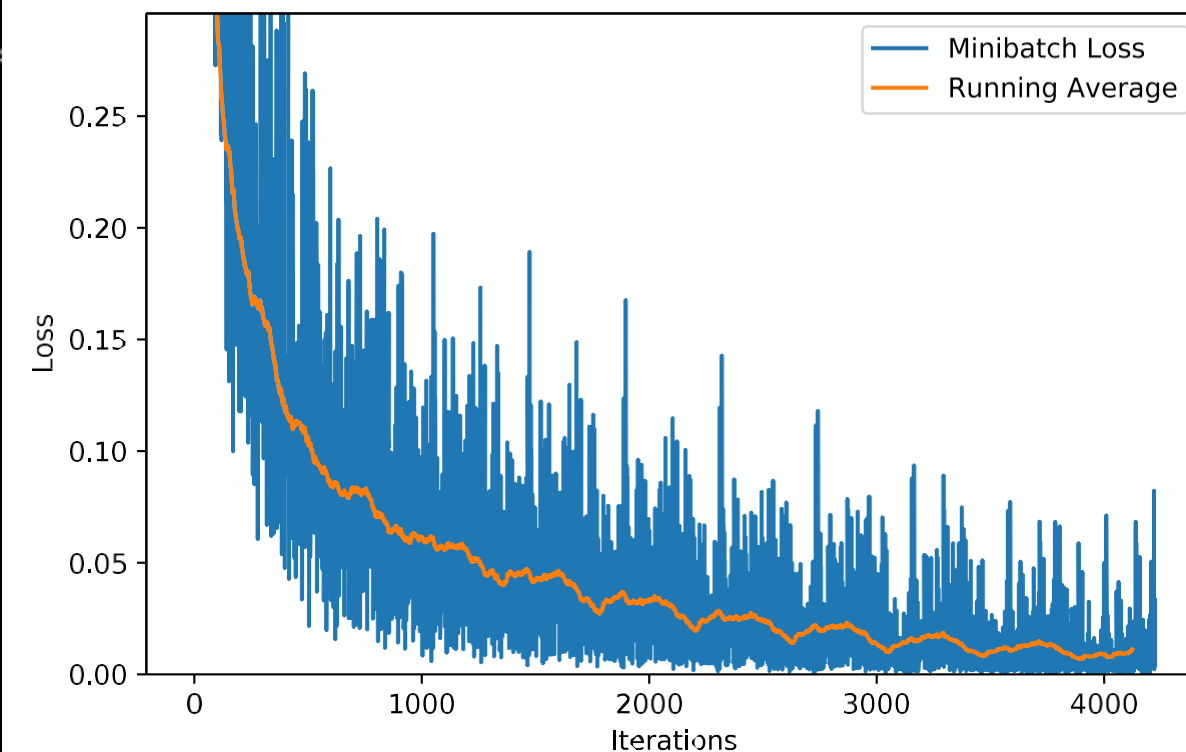


<https://code.visualstudio.com>

<https://github.com/rasbt/stat453-deep-learning-ss21/tree/main/L01/code>


```
(base) raschka@lambda-quad:~/code/stat453-ss21-exp$ python simple_cnn.py
PyTorch version: 1.7.0
Using cuda:0
[W Context.cpp:69] Warning: torch.set_deterministic is in beta, and its de
on operator())
Epoch: 001/010 | Batch 0000/0422 | Loss: 2.2935
Epoch: 001/010 | Batch 0050/0422 | Loss: 0.5462
Epoch: 001/010 | Batch 0100/0422 | Loss: 0.3154
Epoch: 001/010 | Batch 0150/0422 | Loss: 0.2551
Epoch: 001/010 | Batch 0200/0422 | Loss: 0.1792
Epoch: 001/010 | Batch 0250/0422 | Loss: 0.2210
Epoch: 001/010 | Batch 0300/0422 | Loss: 0.1551
Epoch: 001/010 | Batch 0350/0422 | Loss: 0.2155
Epoch: 001/010 | Batch 0400/0422 | Loss: 0.2306
Epoch: 001/010 | Train: 96.72% | Validation: 97.15%
Time elapsed: 0.09 min
Epoch: 002/010 | Batch 0000/0422 | Loss: 0.1028
Epoch: 002/010 | Batch 0050/0422 | Loss: 0.1167
Epoch: 002/010 | Batch 0100/0422 | Loss: 0.0660
Epoch: 002/010 | Batch 0150/0422 | Loss: 0.1024
Epoch: 002/010 | Batch 0200/0422 | Loss: 0.0847
Epoch: 002/010 | Batch 0250/0422 | Loss: 0.0905
Epoch: 002/010 | Batch 0300/0422 | Loss: 0.1024
Epoch: 002/010 | Batch 0350/0422 | Loss: 0.0719
Epoch: 002/010 | Batch 0400/0422 | Loss: 0.1302
Epoch: 002/010 | Train: 98.12% | Validation: 98.03%
Time elapsed: 0.18 min
Epoch: 003/010 | Batch 0000/0422 | Loss: 0.0720
Epoch: 003/010 | Batch 0050/0422 | Loss: 0.0984
Epoch: 003/010 | Batch 0100/0422 | Loss: 0.0373
Epoch: 003/010 | Batch 0150/0422 | Loss: 0.0685
Epoch: 003/010 | Batch 0200/0422 | Loss: 0.0511
Epoch: 003/010 | Batch 0250/0422 | Loss: 0.0617
Epoch: 003/010 | Batch 0300/0422 | Loss: 0.0748
```

```
Epoch: 010/010 | Batch 0000/0422 | Loss: 0.0095
Epoch: 010/010 | Batch 0050/0422 | Loss: 0.0113
Epoch: 010/010 | Batch 0100/0422 | Loss: 0.0135
Epoch: 010/010 | Batch 0150/0422 | Loss: 0.0028
Epoch: 010/010 | Batch 0200/0422 | Loss: 0.0019
Epoch: 010/010 | Batch 0250/0422 | Loss: 0.0049
Epoch: 010/010 | Batch 0300/0422 | Loss: 0.0132
Epoch: 010/010 | Batch 0350/0422 | Loss: 0.0114
Epoch: 010/010 | Batch 0400/0422 | Loss: 0.0270
Epoch: 010/010 | Train: 99.55% | Validation: 98.68%
Time elapsed: 0.88 min
Total Training Time: 0.88 min
Test accuracy 98.66%
```



Next Lecture:

A Brief Summary of the
History of
Neural Networks and Deep Learning